



# Light as a Versatile Platform for Analogue Physical Simulations and Quantum Imaging Applications

Armchair: Tiago D. Ferreira

Workshop: Quantum-SOPHA and Armchairs, Peyresq, 2026



**INESCTEC**



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Light as a Versatile Platform for **Analogue Physical Simulations** and Quantum Imaging Applications

# Analogue Physical Simulators of Quantum Fluids

Quantum fluids are **hydrodynamical systems** that **exhibit quantum effects**, such as superfluidity or quantum turbulence.

Quantum fluids of light tries to **emulate these systems** with **light**.

# Analogue Physical Simulators of Quantum Fluids

Systems governed by a similar mathematical model

Quantum Fluid

$$\left[ i\hbar \frac{\partial \psi(\vec{r}_\perp, t)}{\partial t} + \left[ \frac{\hbar^2}{2m} \nabla_\perp^2 + V_{ext}(\vec{r}_\perp, t) + g |\psi(\vec{r}_\perp, t)|^2 + i\gamma \right] \right] \psi(\vec{r}_\perp, t) = 0 \rightarrow$$

Time      Mass      External potential      Interactions      Losses

Vortex nucleation due to a defect



They **support macroscopic quantum effects**.

# Analogue Physical Simulators of Quantum Fluids

Systems governed by a similar mathematical model

Quantum Fluid

$$\left[ i\hbar \frac{\partial \psi(\vec{r}_\perp, t)}{\partial t} + \left[ \frac{\hbar^2}{2m} \nabla_\perp^2 + V_{ext}(\vec{r}_\perp, t) + g |\psi(\vec{r}_\perp, t)|^2 + i\gamma \right] \right] \psi(\vec{r}_\perp, t) = 0 \rightarrow$$

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Vortex nucleation due to a defect

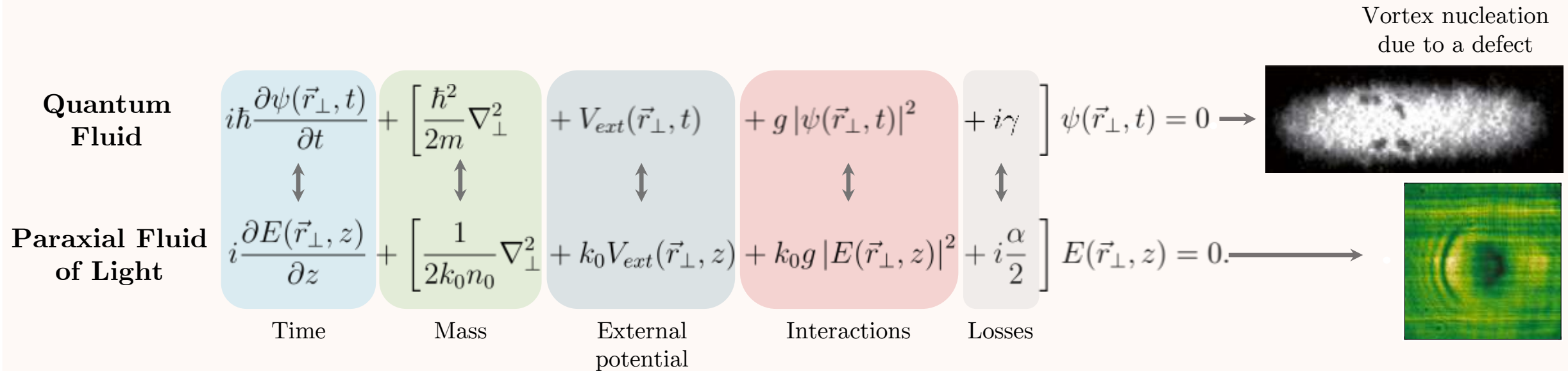


**However**, they require **low-temperatures** and **expensive experimental setups**.

**Accessing** all the **fluid properties**, such as the velocity field, is **difficult**.

# Analogue Physical Simulators of Quantum Fluids

Systems governed by a similar mathematical model



Paraxial fluids of light are **suitable physical system** capable of **emulating quantum fluids**.



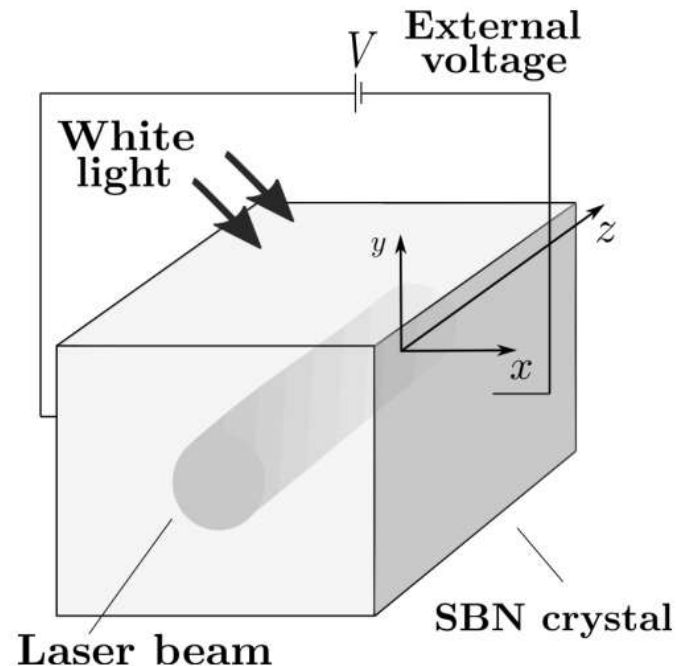
# Paraxial Fluids of Light at INESC TEC

At **INESC TEC** we use **Photorefractive Crystals** as the nonlinear medium for paraxial fluids of light

A **light beam propagating** in a **photorefractive crystal** can be described by the following model:

$$i\frac{\partial E_f}{\partial z} + \frac{1}{2k_f n_e} \nabla_{\perp}^2 E_f + k_f \Delta n_{max} \frac{|E_f|^2}{|E_f|^2 + I_{sat}} E_f + i\frac{\alpha}{2} E_f = 0$$

**Figure 1** – Photorefractive crystal. Image adapted from “Fluids of light in a nonlinear photorefractive medium”, Omar Boughdad, 2020.



$$\Delta n_{max} \propto V_{Ext}$$

$$I_{sat} \propto \text{White light}$$

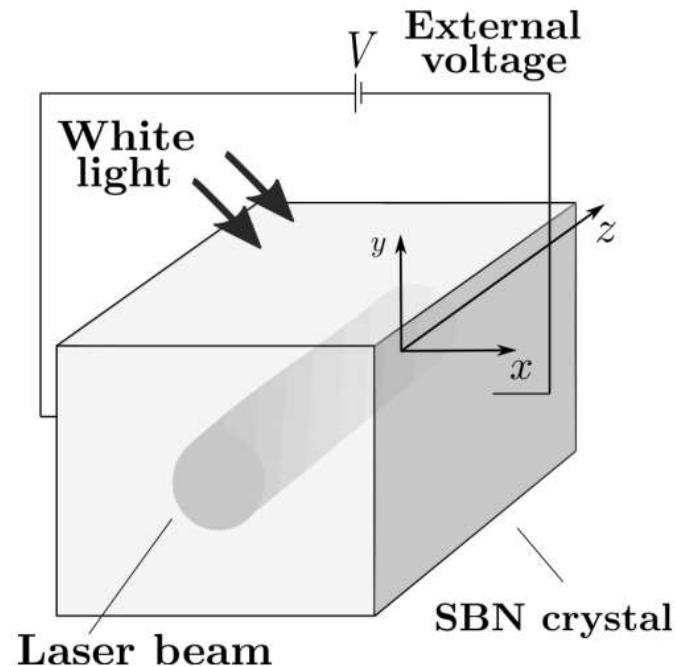
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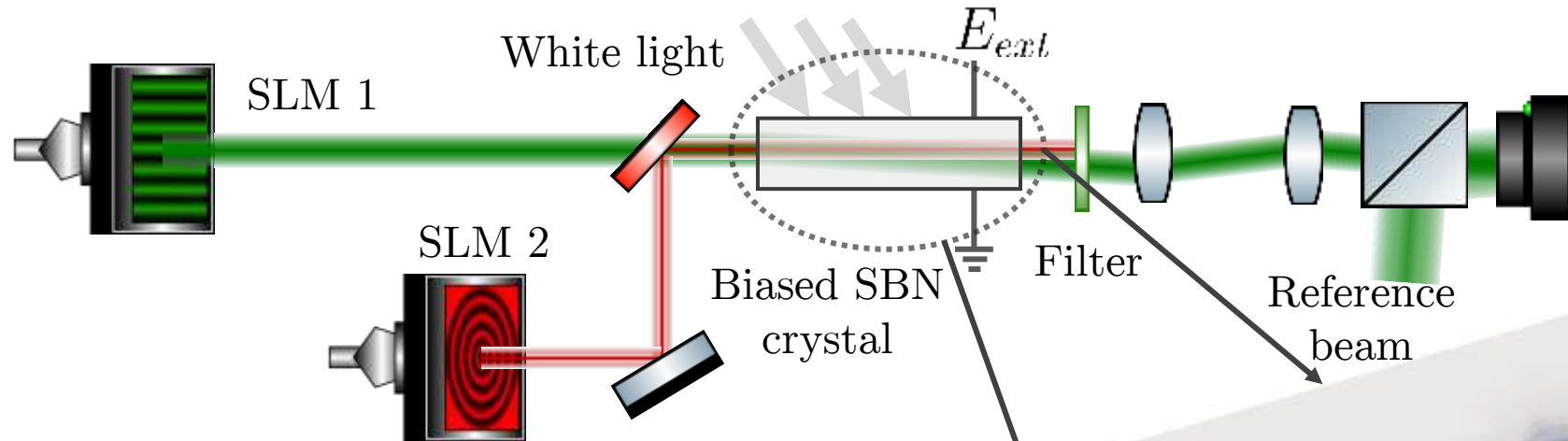


- ✓ Low power ( $\sim \mu W$ )
- ✓ Cheaper to assemble
- ✗ Anisotropic and slow response

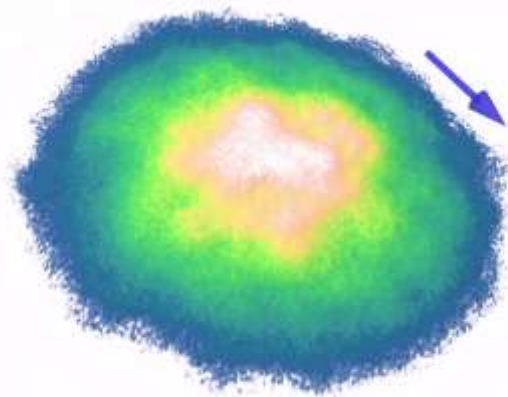
Experimental work started in 2022.

# Applications: Analogue Quantum Turbulence

## Experimental setup

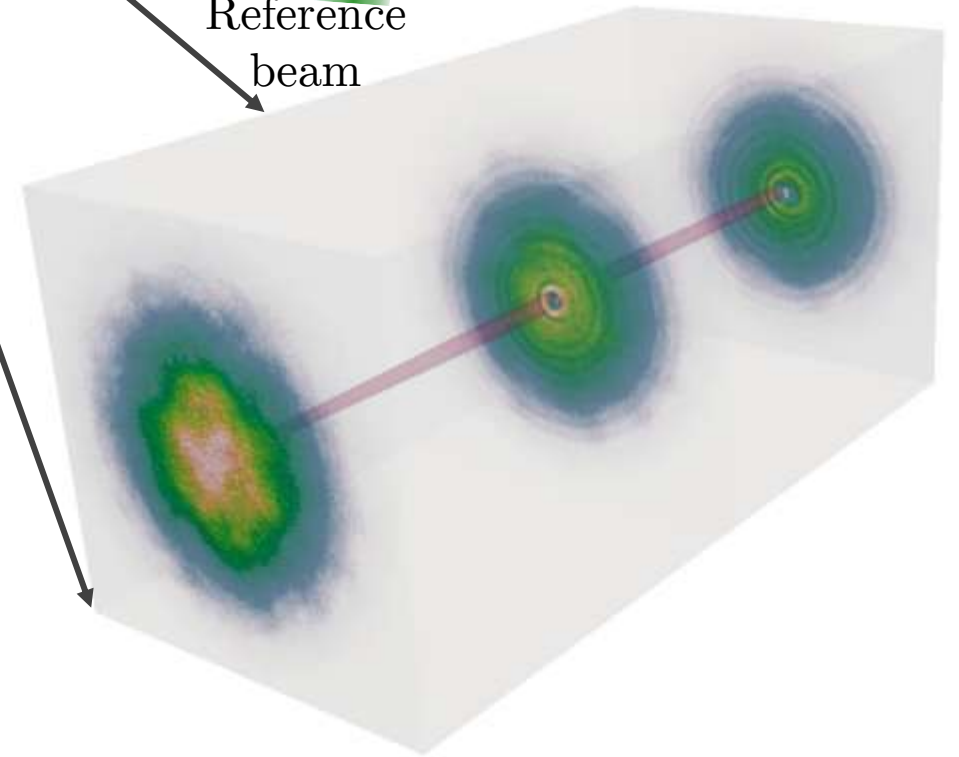


**Figure 2** – Experimental setup.



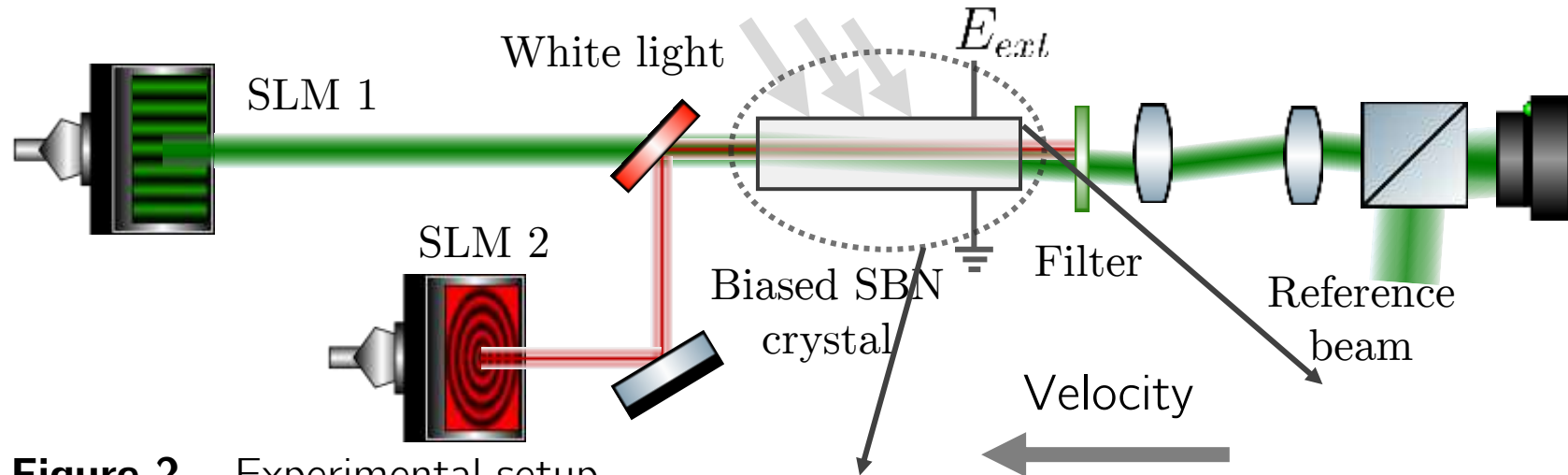
**Figure 3** – Time evolution of the 2D fluid of light.

**Figure 4** – 3D representation of the dynamics.



# Applications: Analogue Quantum Turbulence

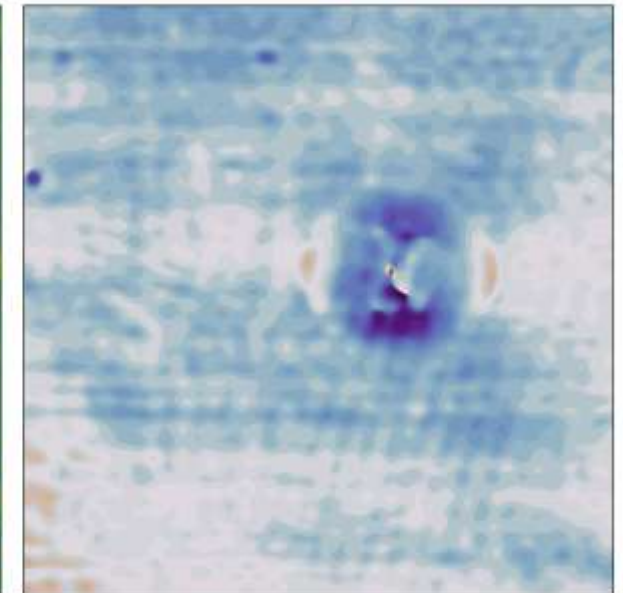
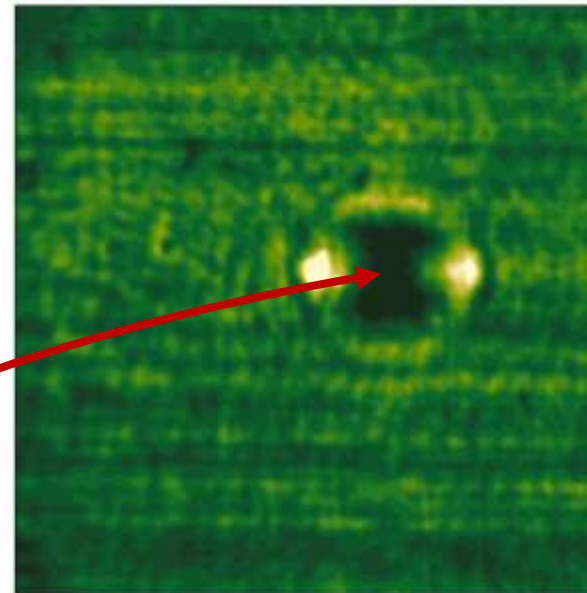
## Experimental setup



**Figure 2** – Experimental setup.

**Figure 3** – Fluid of Light evolution for different velocities. The defect is not visible.

Optical  
Defect

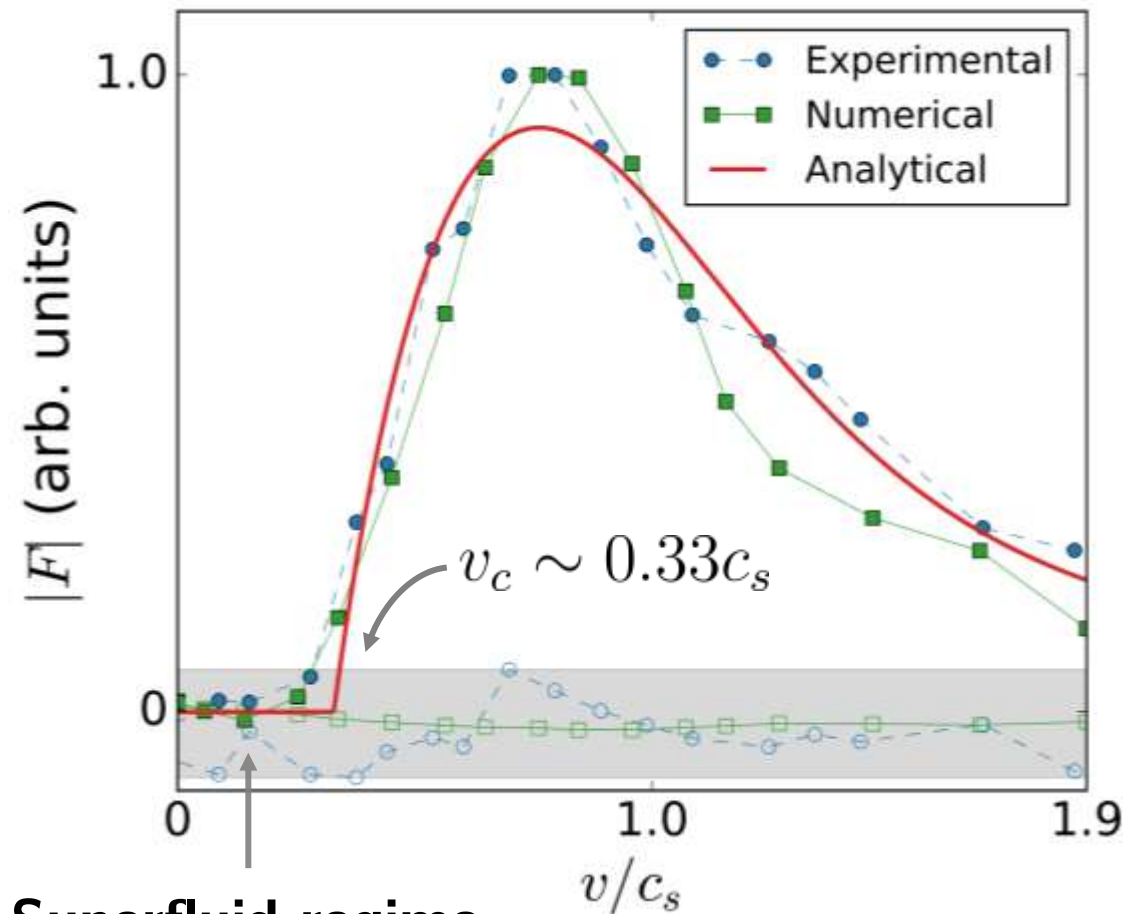


**Figure 15** – 3D representation of the dynamics.

# Applications: Analogue Quantum Turbulence

Experimental results: Characterizing the turbulent states

**Figure 4** – Drag-Force.

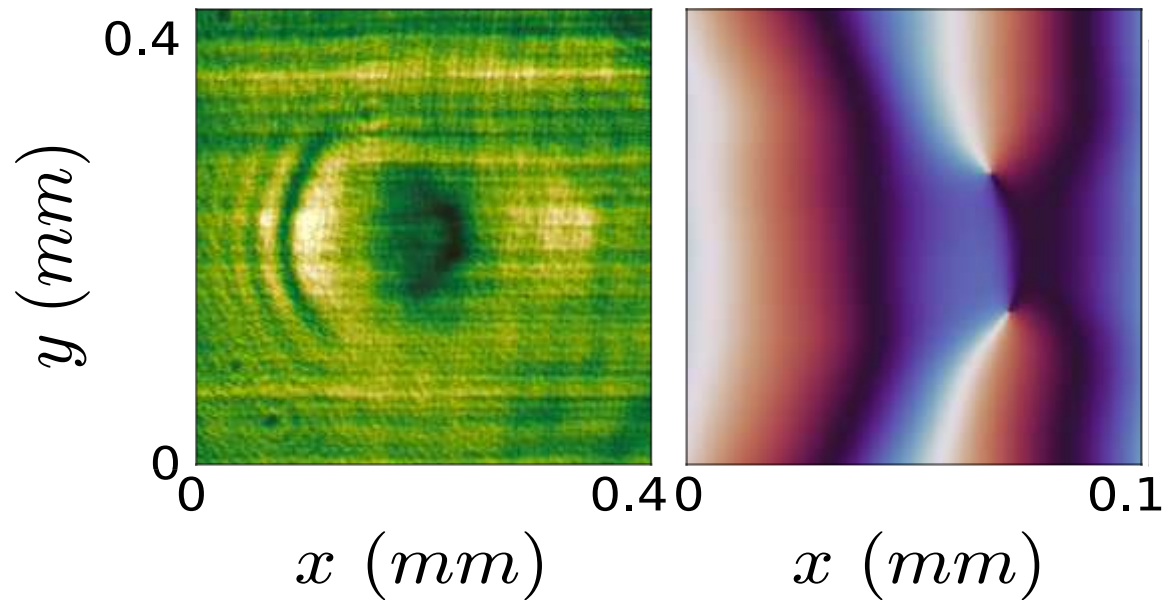


$$\begin{aligned}\vec{F} &= - \int |\psi_f|^2 \nabla V(\vec{r}) d\vec{r}_\perp \\ &= \begin{cases} F // v \\ F \perp v \end{cases}\end{aligned}$$

The force has two components:  
**Parallel** to the **velocity**  
**Perpendicular** to the **velocity**

# Applications: Analogue Quantum Turbulence

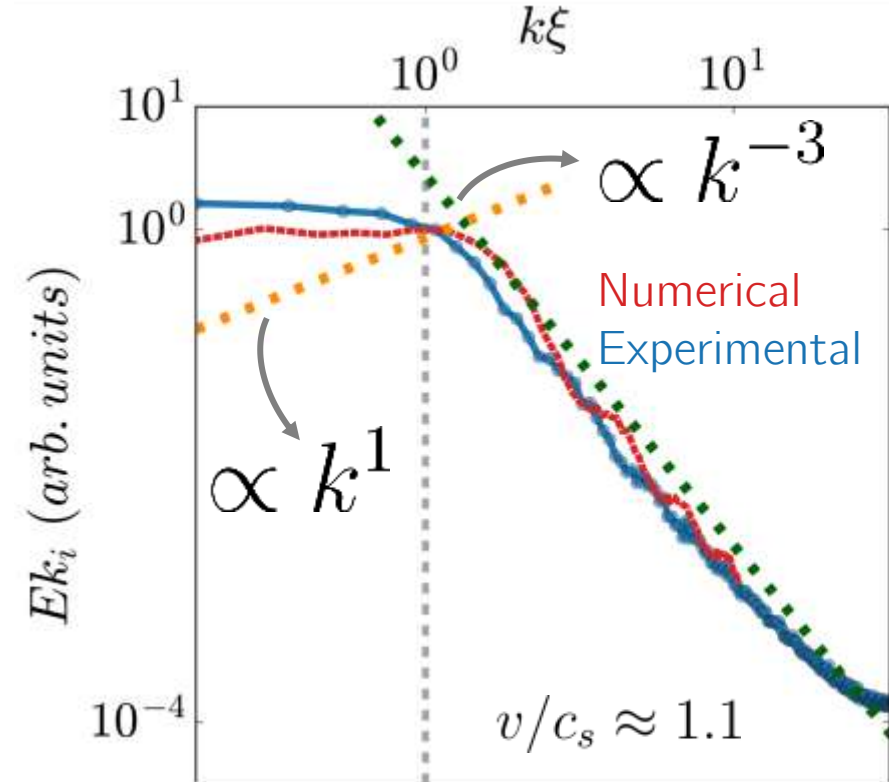
Experimental results: Characterizing the turbulent states



**Figure 5** – Fluid of light at the output.

Created **turbulent states** with **quantum vortices!**

$k^{-3} \longrightarrow$  In **quantum** fluids.



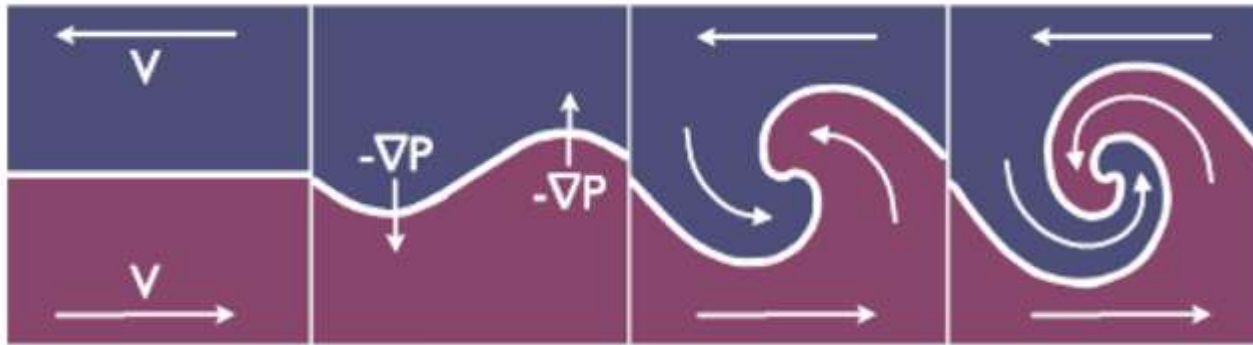
**Figure 6** – Incompressible component of the kinetic energy.



# Applications: Kelvin-Helmholtz Instability in Fluids of Light

The Kelvin-Helmholtz Instability **appears** at the **interface between two fluids** with **different velocities**.

**Vortex structures** are **formed** at the **interface**.



**Figure 7** – Formation of the Kelvin-Helmholtz Instability in classical fluids. Image adapted from 10.1007/s11214-014-0085-z.

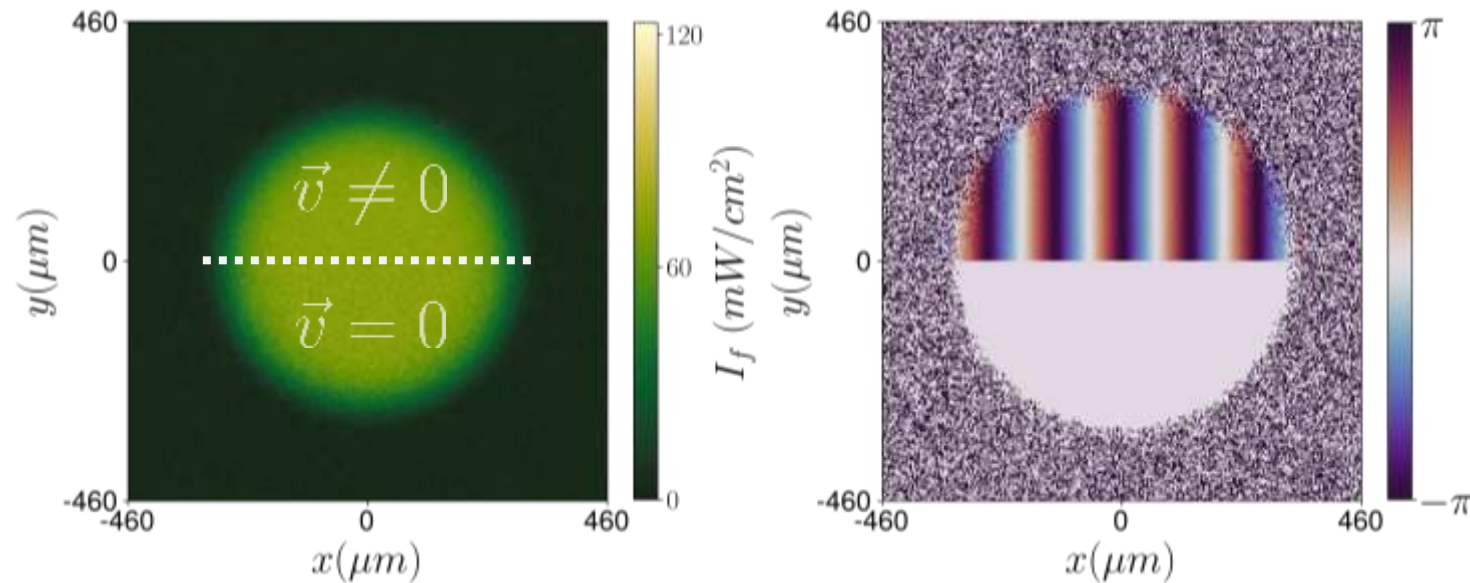


**Figure 8** – Kelvin-Helmholtz Instability clouds. Image adapted from Brooks Martner, NOAA/ETL

# Applications: Kelvin-Helmholtz Instability in Fluids of Light

We can **emulate** a **similar system** with **fluids of light**.

By considering a **single beam**, where **part** of the **beam** has a **velocity**, and the **remaining** is **stationary**.

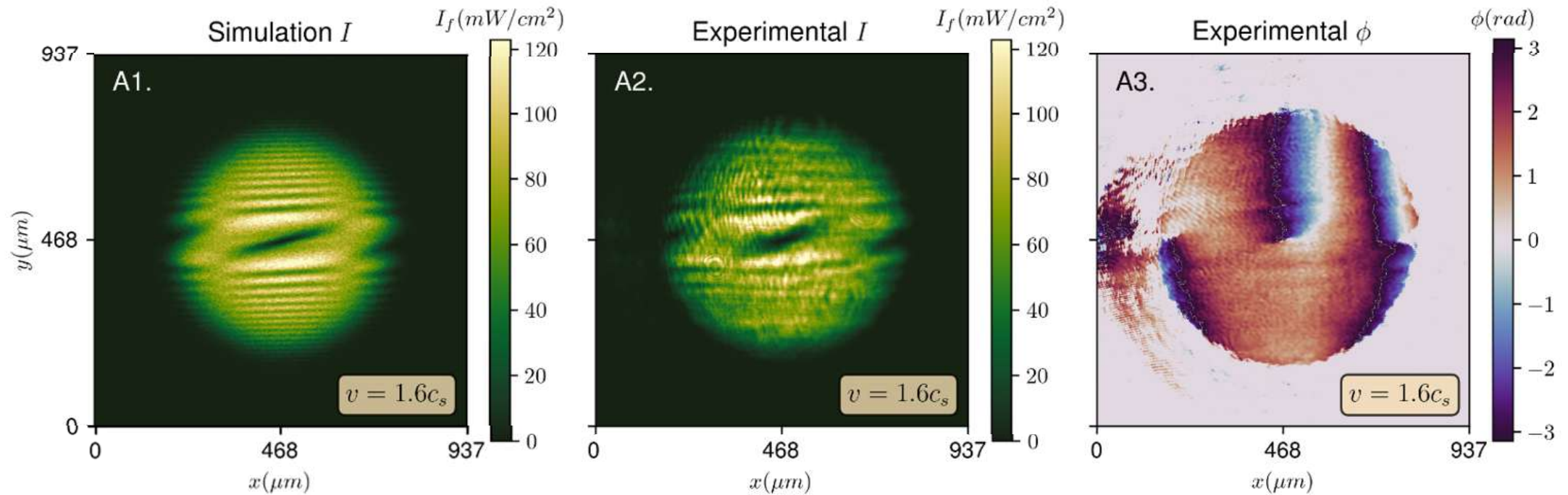


**Figure 9** – Initial state for observing the Kelvin-Helmholtz Instability.



# Applications: Kelvin-Helmholtz Instability in Fluids of Light

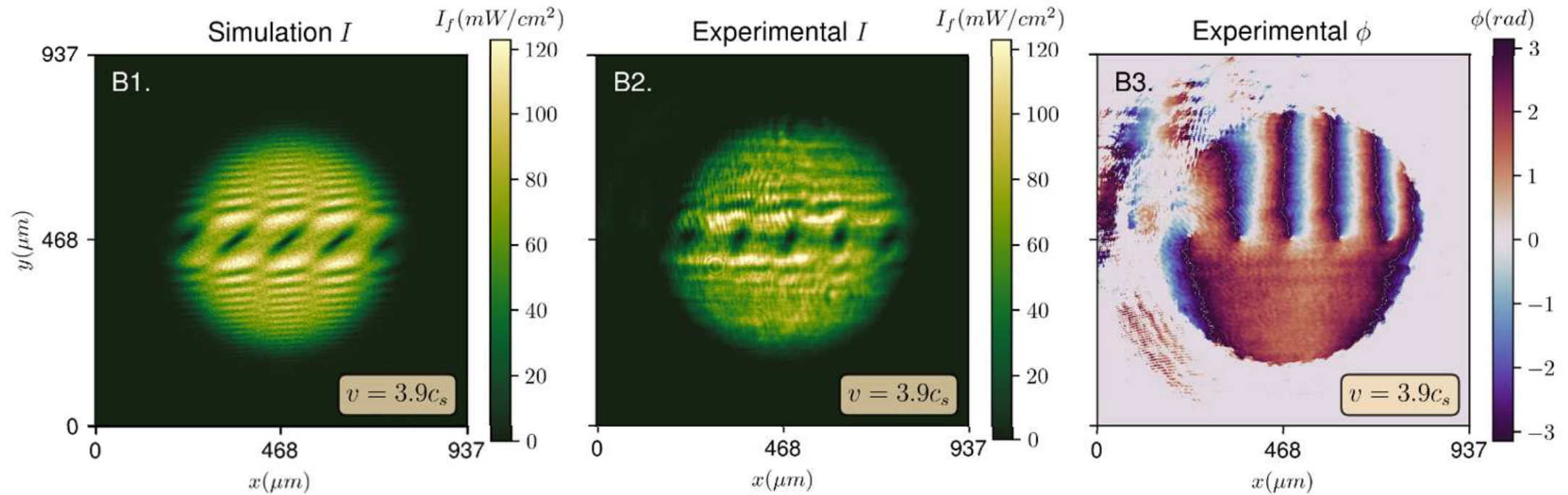
Experimental results: vortex generation at the interface



**Figure 10** – Profiles of the output states for the Kelvin-Helmholtz Instability with different velocities.

# Applications: Kelvin-Helmholtz Instability in Fluids of Light

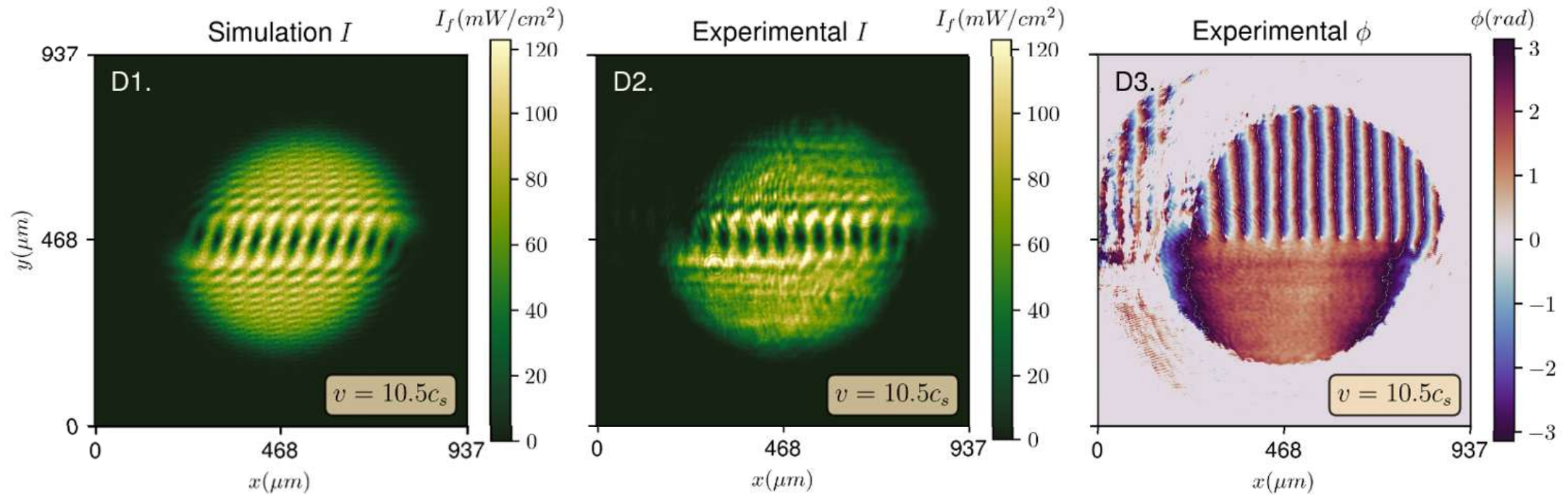
Experimental results: vortex generation at the interface



**Figure 11** – Profiles of the output states for the Kelvin-Helmholtz Instability with different velocities.

# Applications: Kelvin-Helmholtz Instability in Fluids of Light

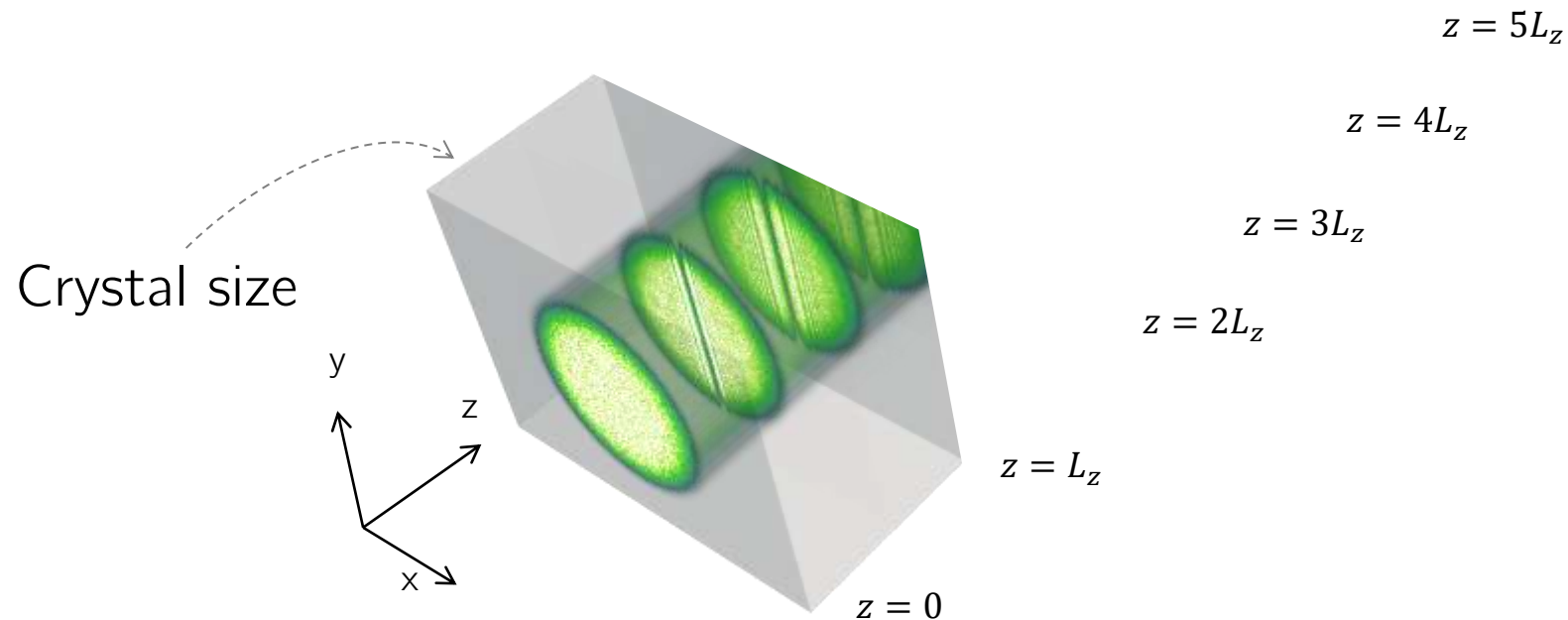
Experimental results: vortex generation at the interface



**Figure 12** – Profiles of the output states for the Kelvin-Helmholtz Instability with different velocities.

# Digital Feedback Loop in Fluids of Light

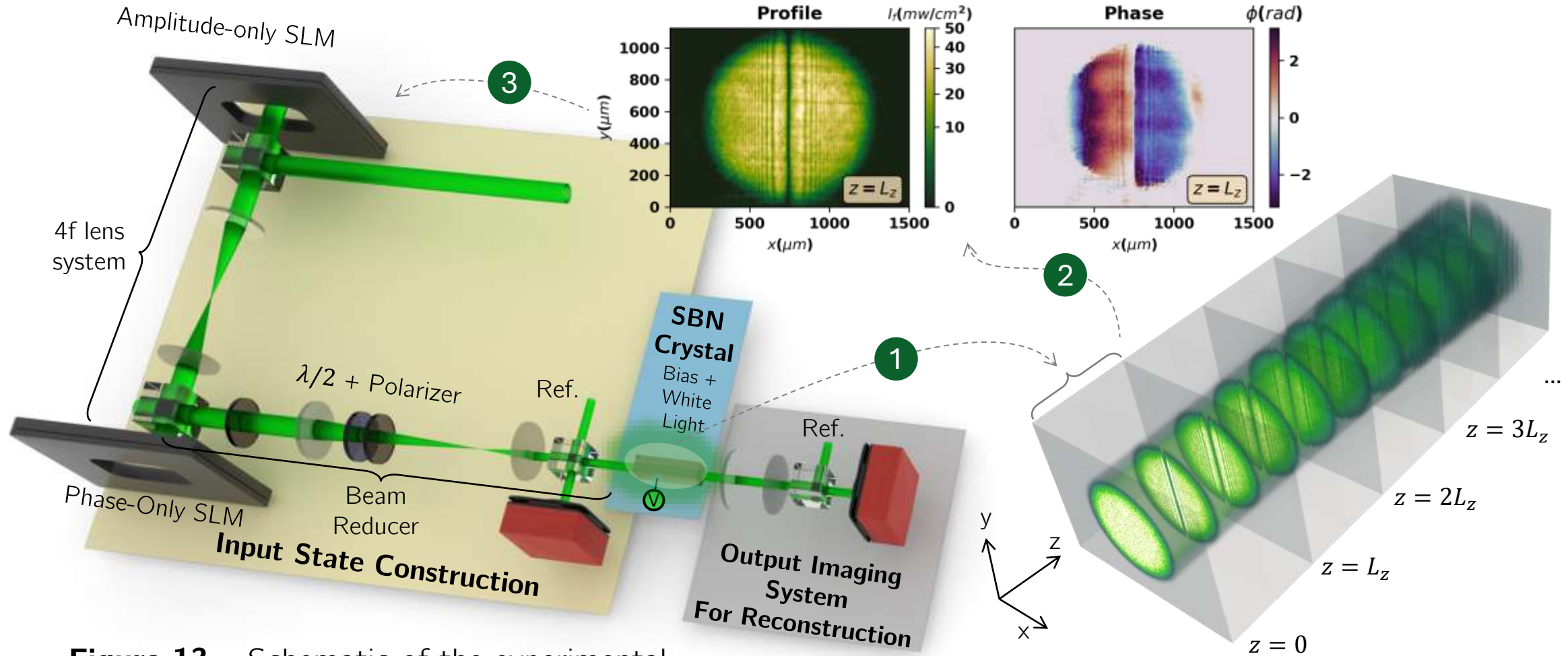
Developed a Digital Feedback Loop to effectively extend the physical size of the nonlinear medium.



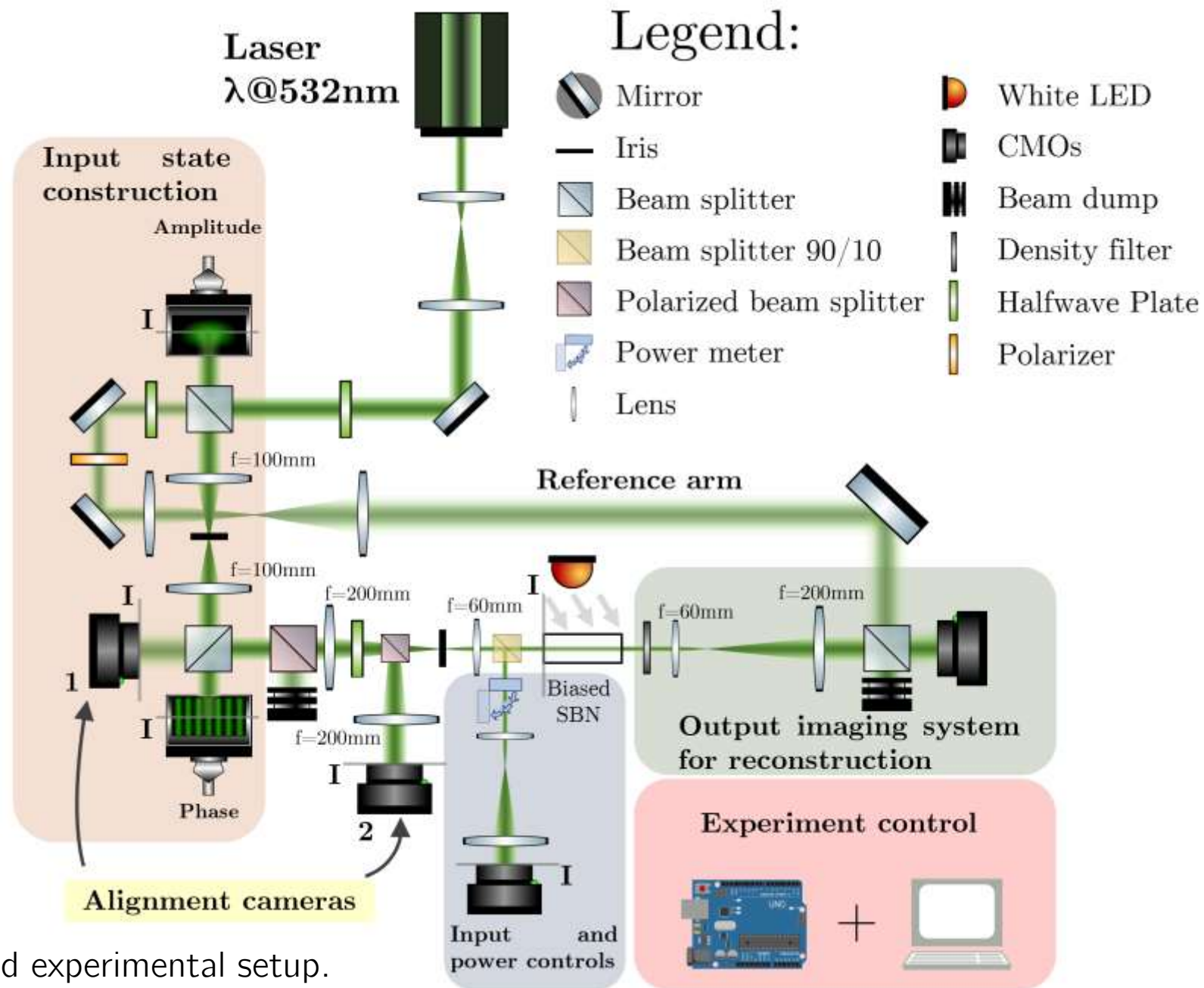


# Digital Feedback Loop in Fluids of Light

## Experimental implementation



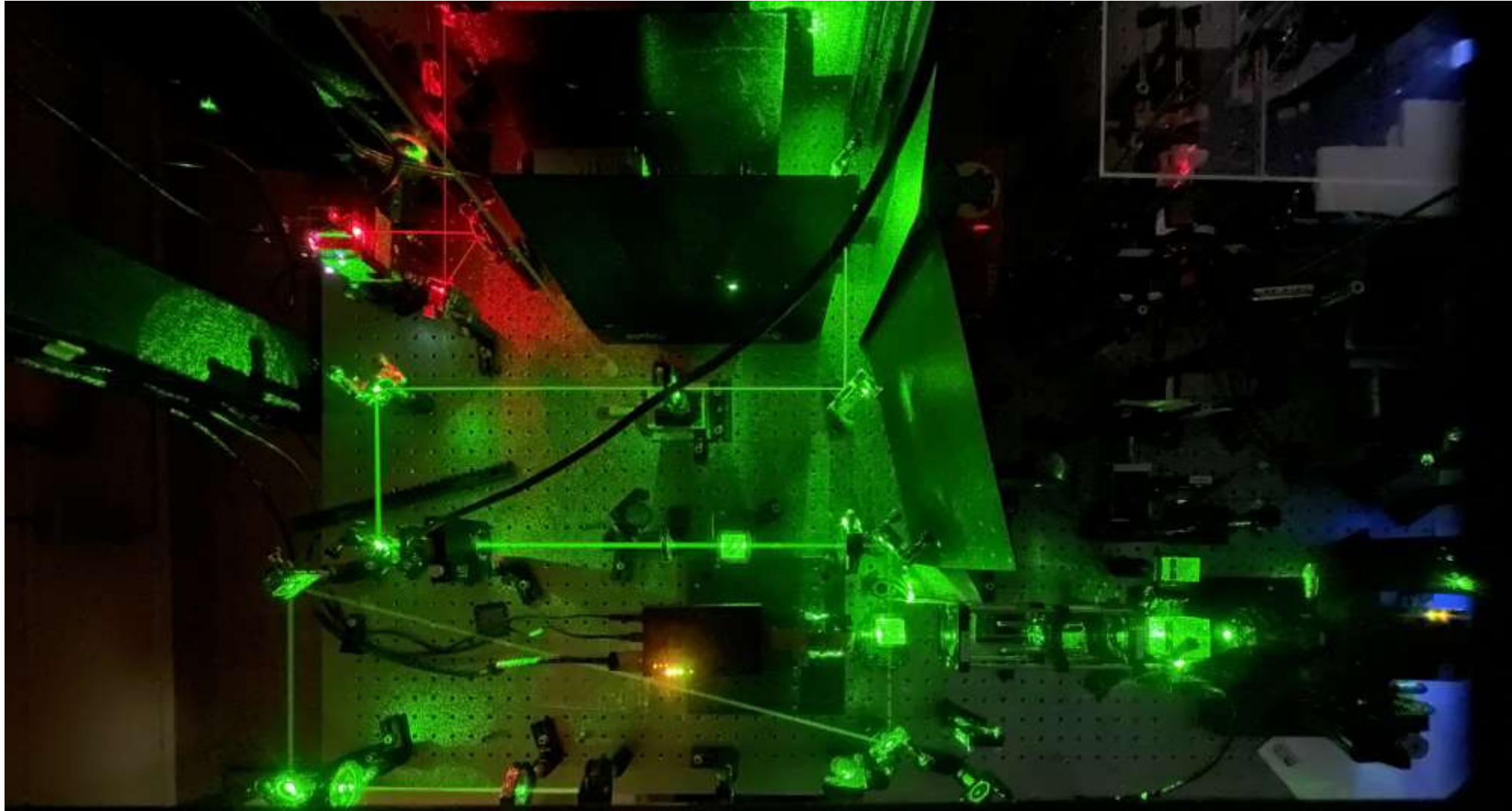
**Figure 13** – Schematic of the experimental scheme used to implement the optical feedback loop.



**Figure 14** – Detailed experimental setup.

# Digital Feedback Loop in Fluids of Light

Experimental implementation



# Digital Feedback Loop in Fluids of Light

Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**

Display Phase and Amplitude



# Digital Feedback Loop in Fluids of Light

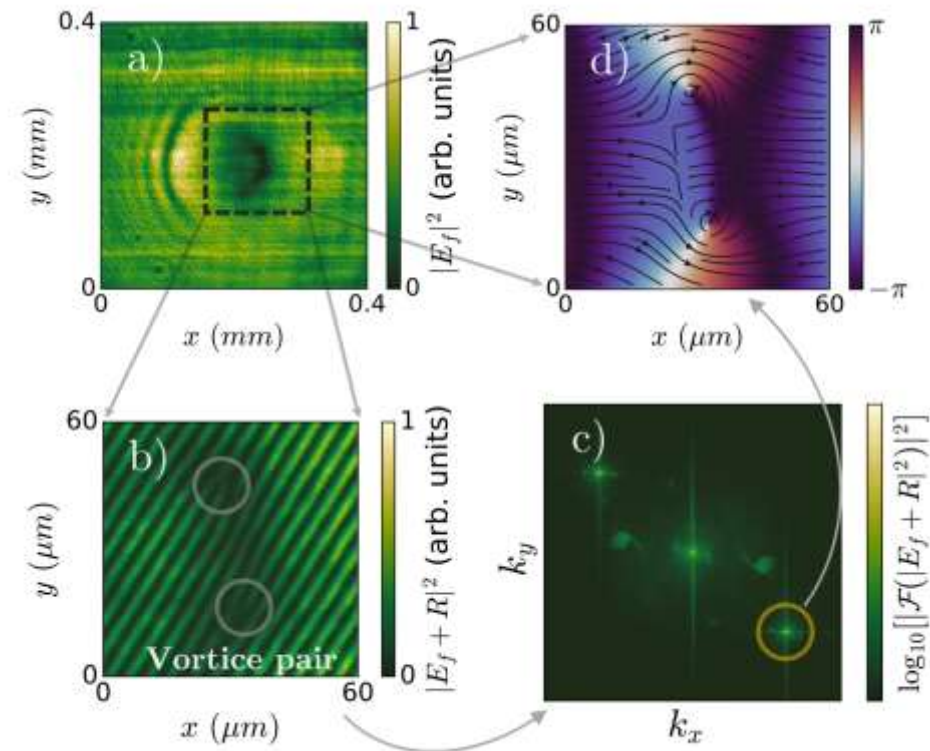
Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**

Display Phase and Amplitude

Acquire complex state at the entrance of the crystal

## Off-axis Digital Holography

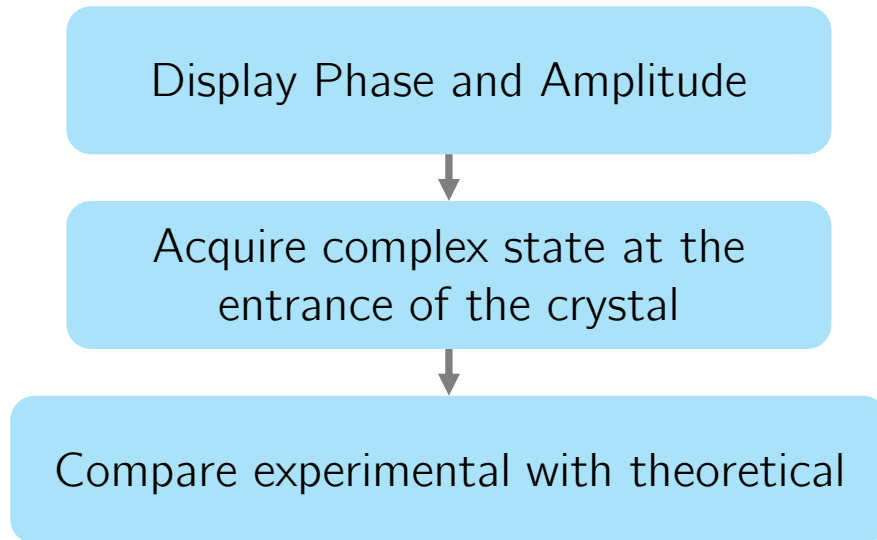


**Figure 15** – Off-axis Digital Holography procedure.

# Digital Feedback Loop in Fluids of Light

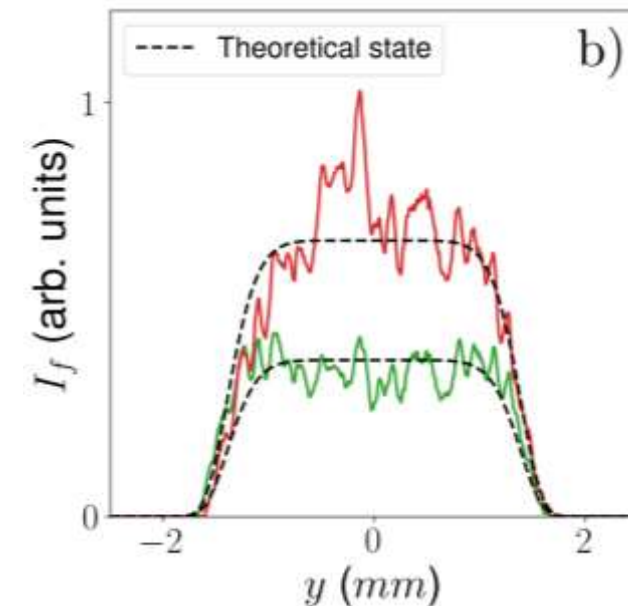
Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**



## Amplitude comparison:

- **Always use** the **same camera exposure**;
- **Good alignment** between the **camera** and **SLM** images.

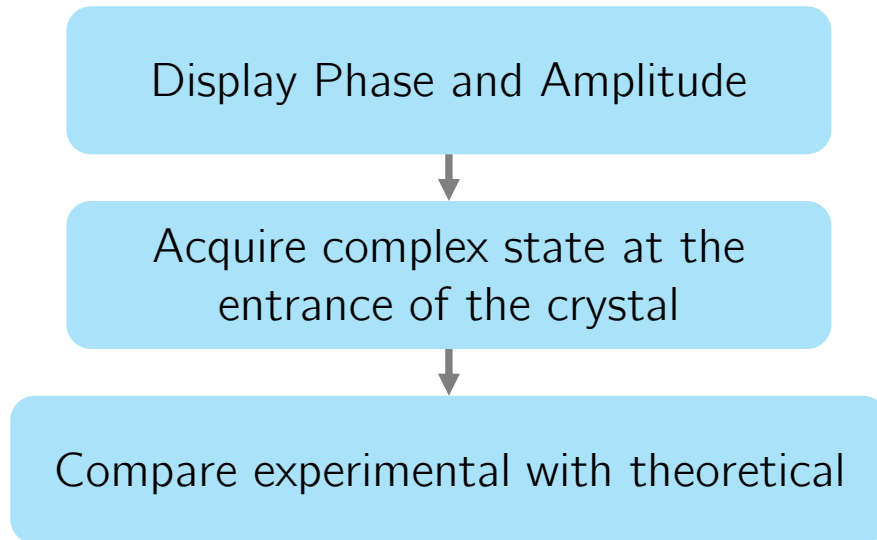


**Figure 16** – Theoretical state vs camera measurement.

# Digital Feedback Loop in Fluids of Light

Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**



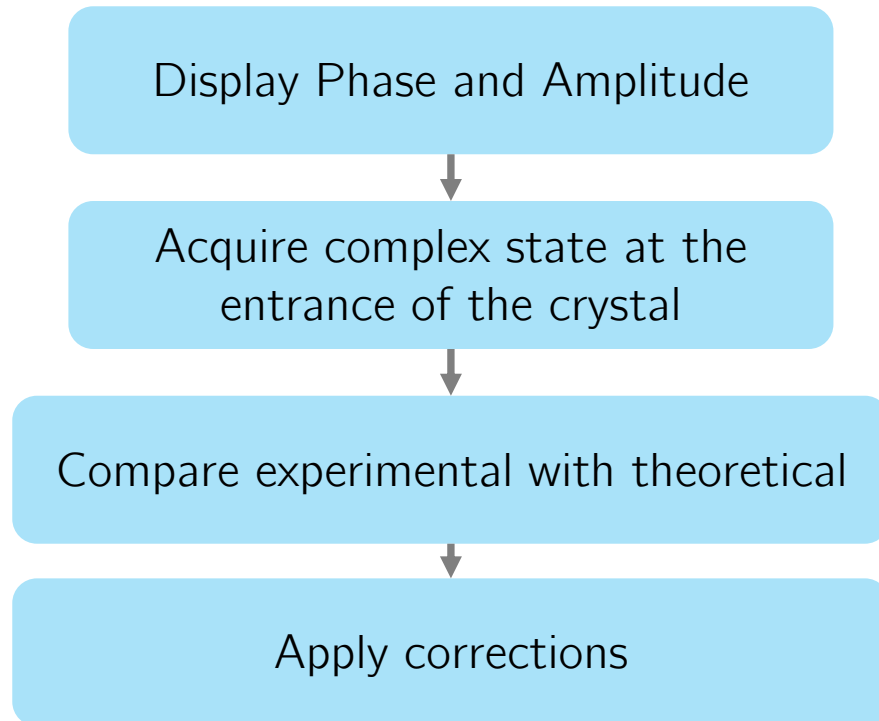
## Phase comparison

- **Experimental** and **theoretical** state **should** have **same global phase**;
- Use **interpolation methods** that **preserve phase singularities**

# Digital Feedback Loop in Fluids of Light

Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**

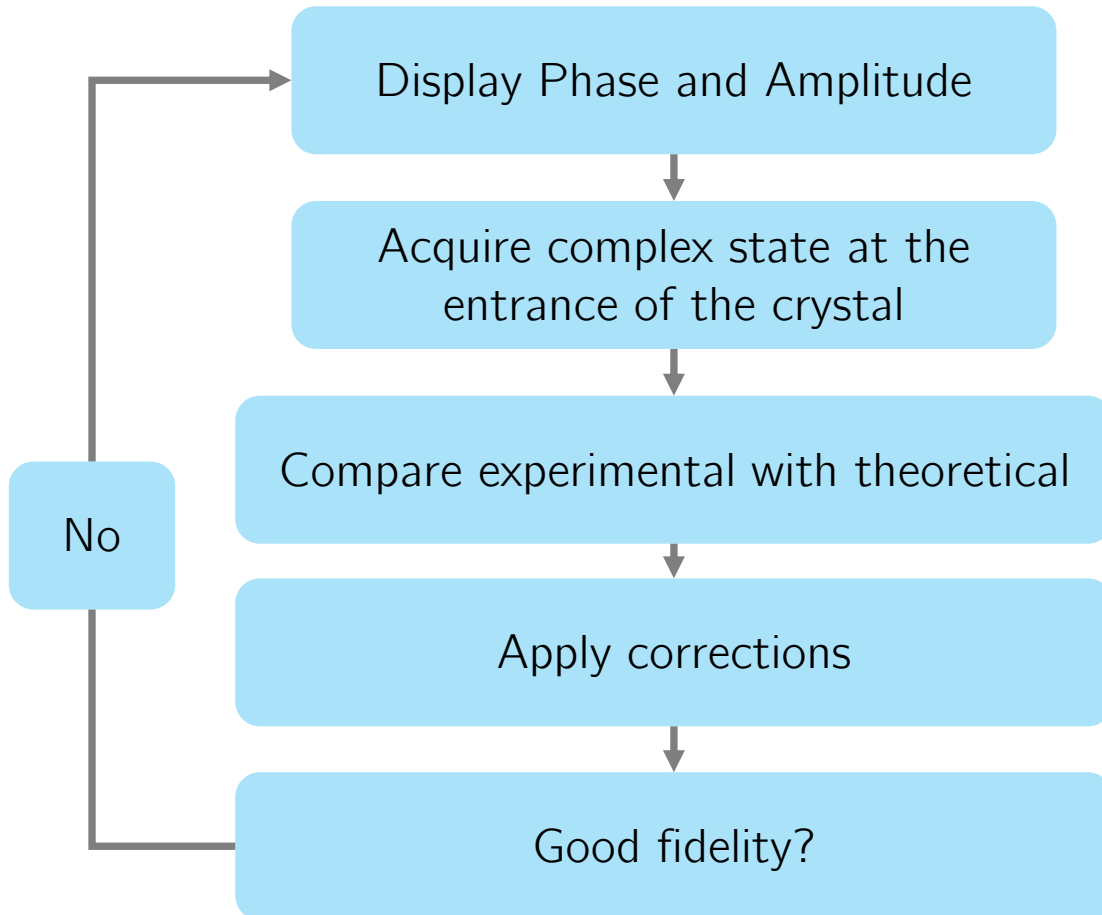


# Digital Feedback Loop in Fluids of Light

Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**

**Repeat the loop**

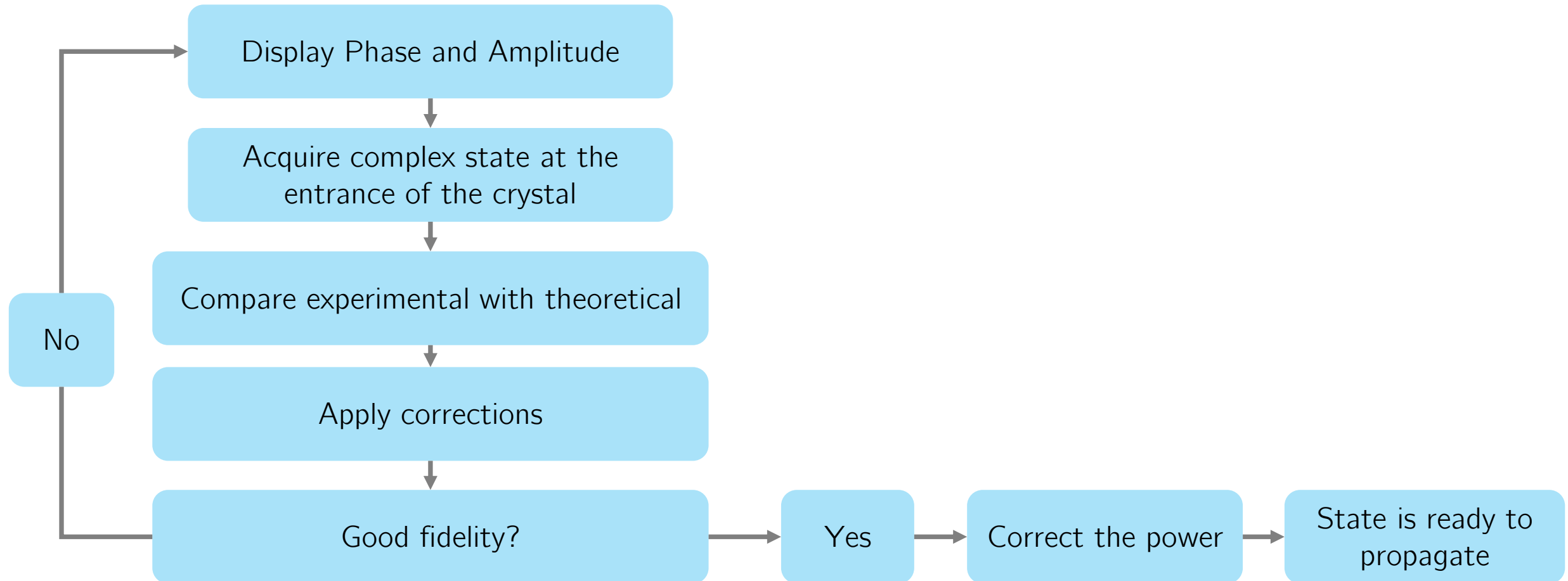


- The **theoretical amplitude** and **phase** profiles are **always** the **initial ones**;
- **Track** the **fidelity** over the **iteration**, and **store** the **state** with the **maximum fidelity**;

# Digital Feedback Loop in Fluids of Light

Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**

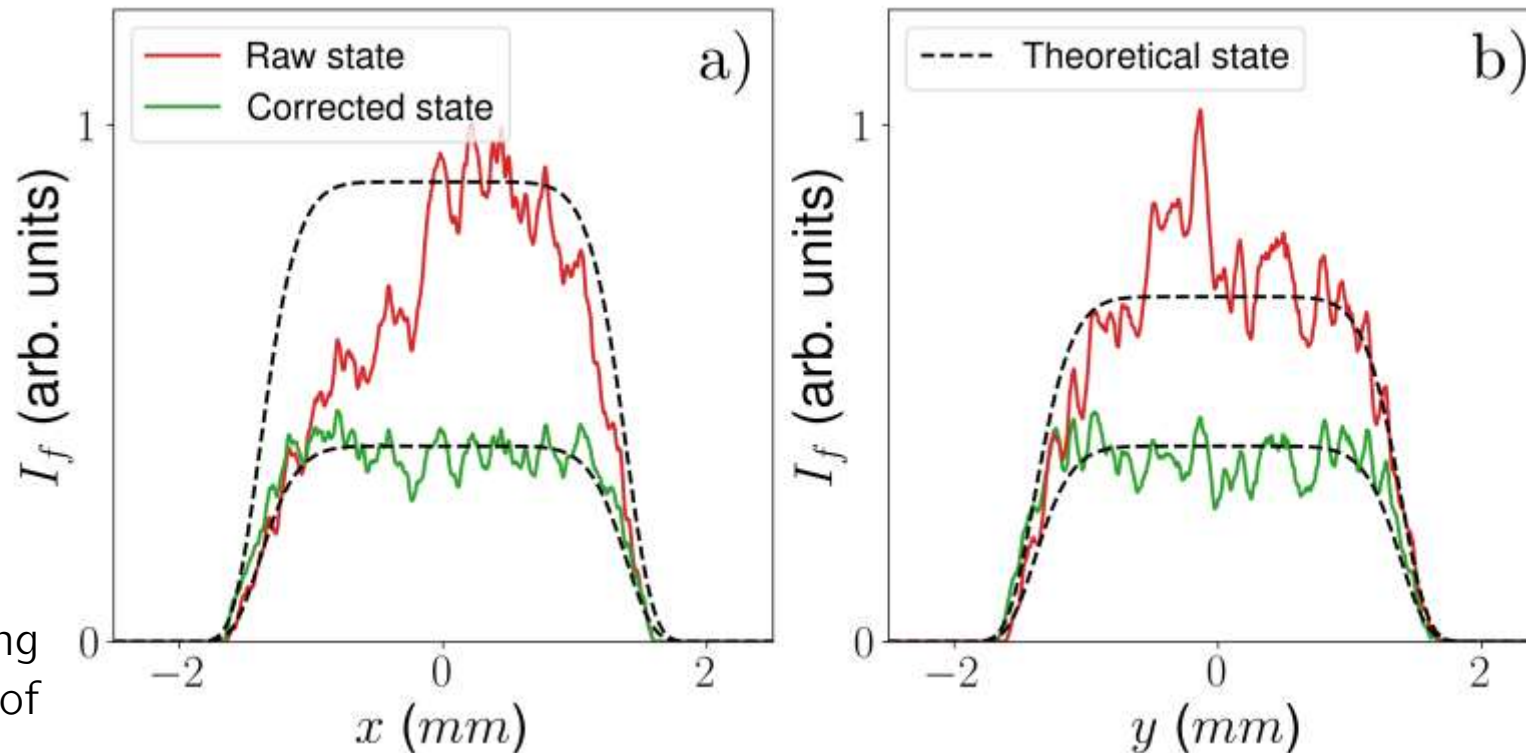


# Digital Feedback Loop in Fluids of Light

Experimental implementation

**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**

We **must compensate** for the usual **Gaussian shape** of the **beam** at the **SLM**



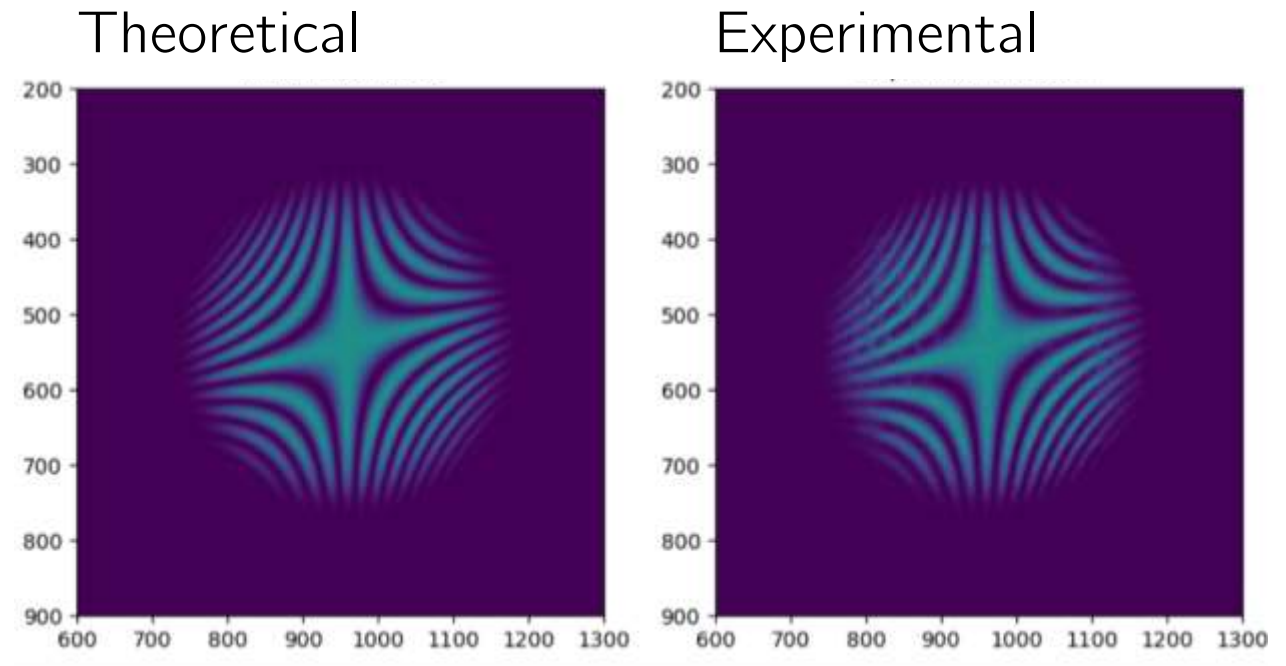
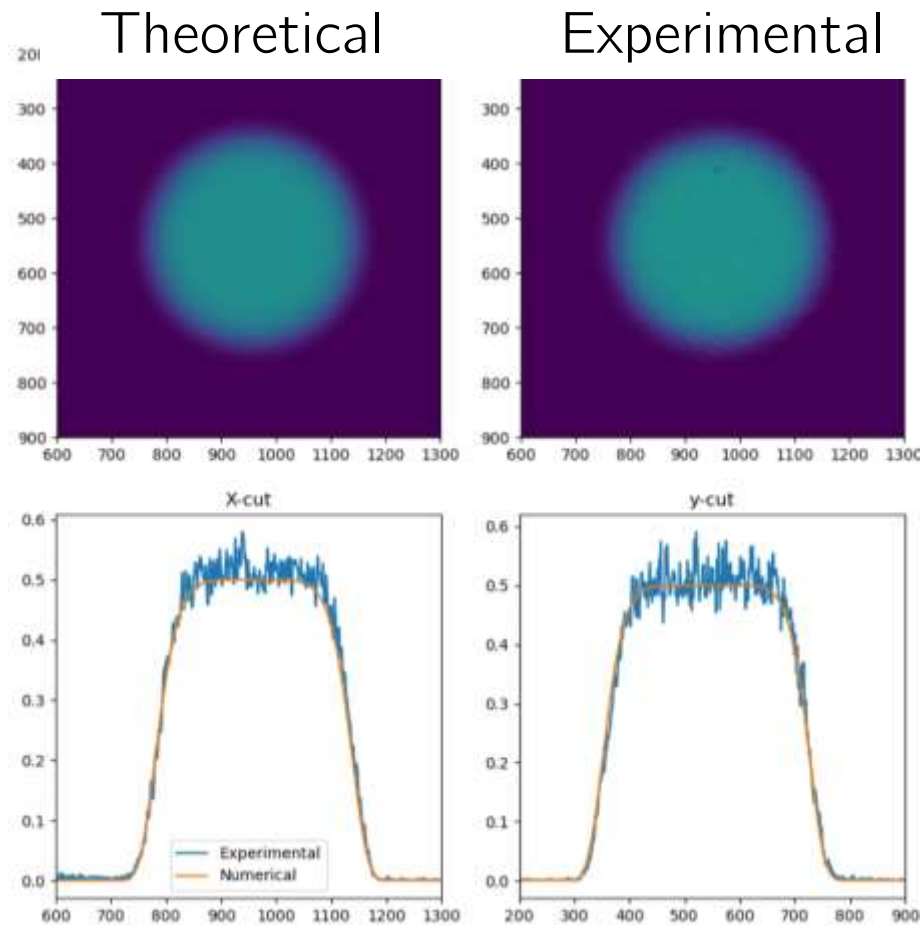
**Figure 17** – Correcting the Gaussian profile of the beam



# Digital Feedback Loop in Fluids of Light

Experimental implementation

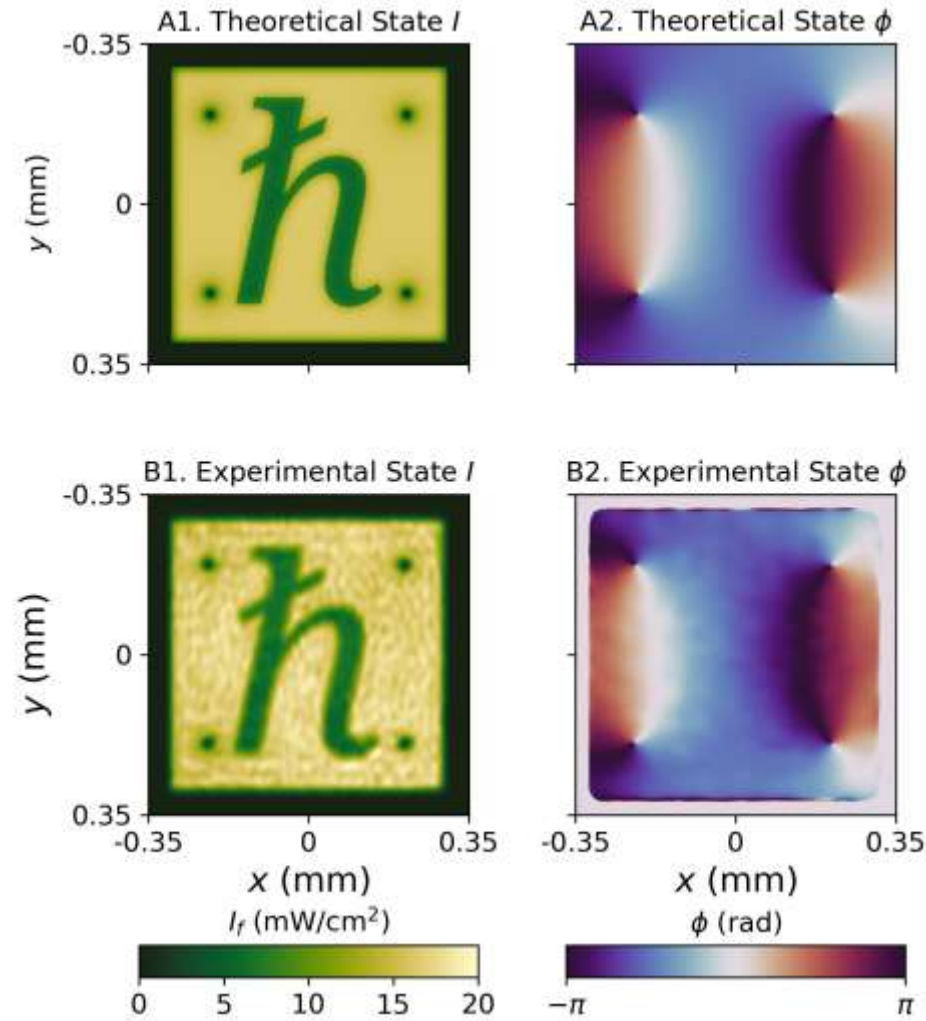
**Optimize** the **amplitude** and **phase** profiles for having state with **high fidelity states**



**Figure 18** – Examples of experimental states with the obtained with the optimization algorithm.

# Digital Feedback Loop in Fluids of Light

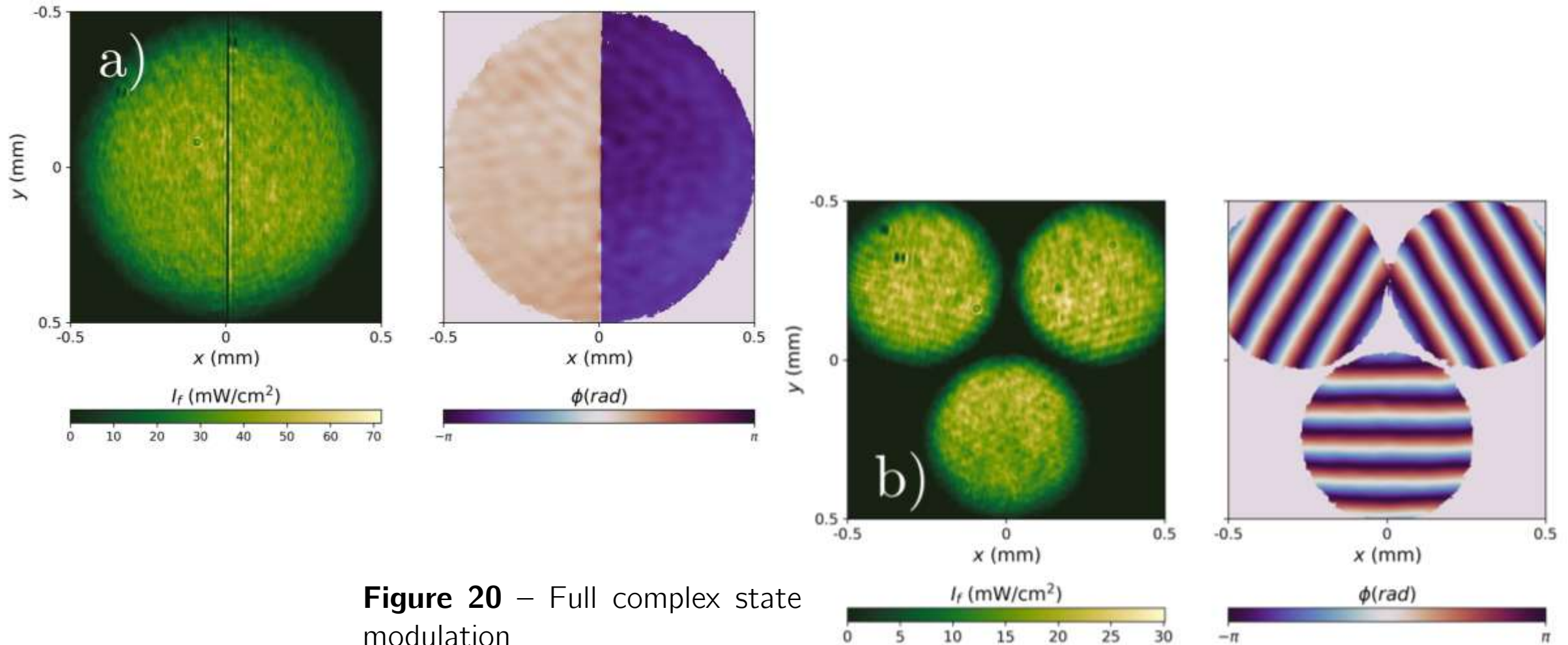
## Experimental implementation



**Figure 19** – Full complex state modulation

# Digital Feedback Loop in Fluids of Light

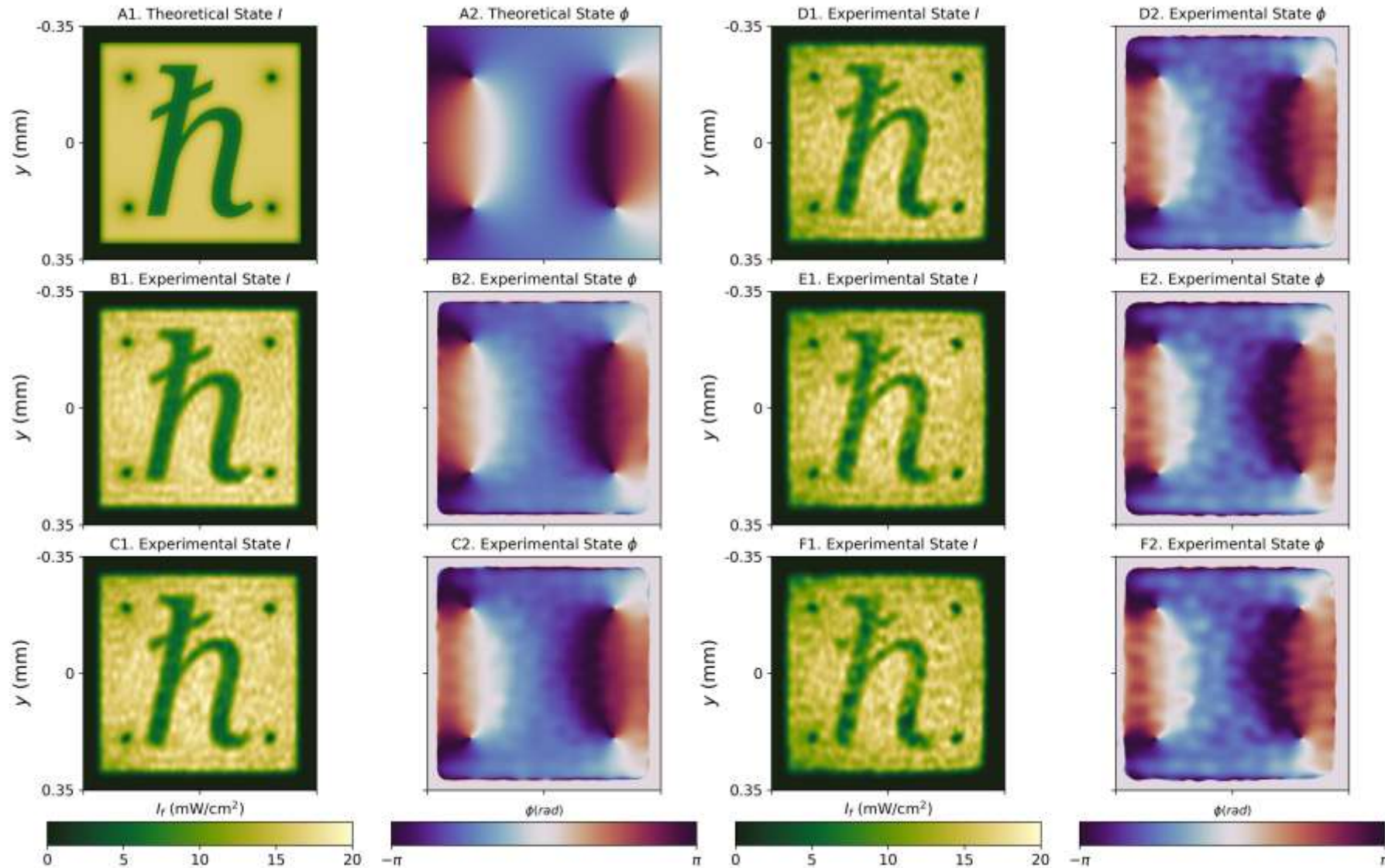
Experimental implementation



**Figure 20** – Full complex state modulation

# Digital Feedback Loop in Fluids of Light

## Experimental implementation



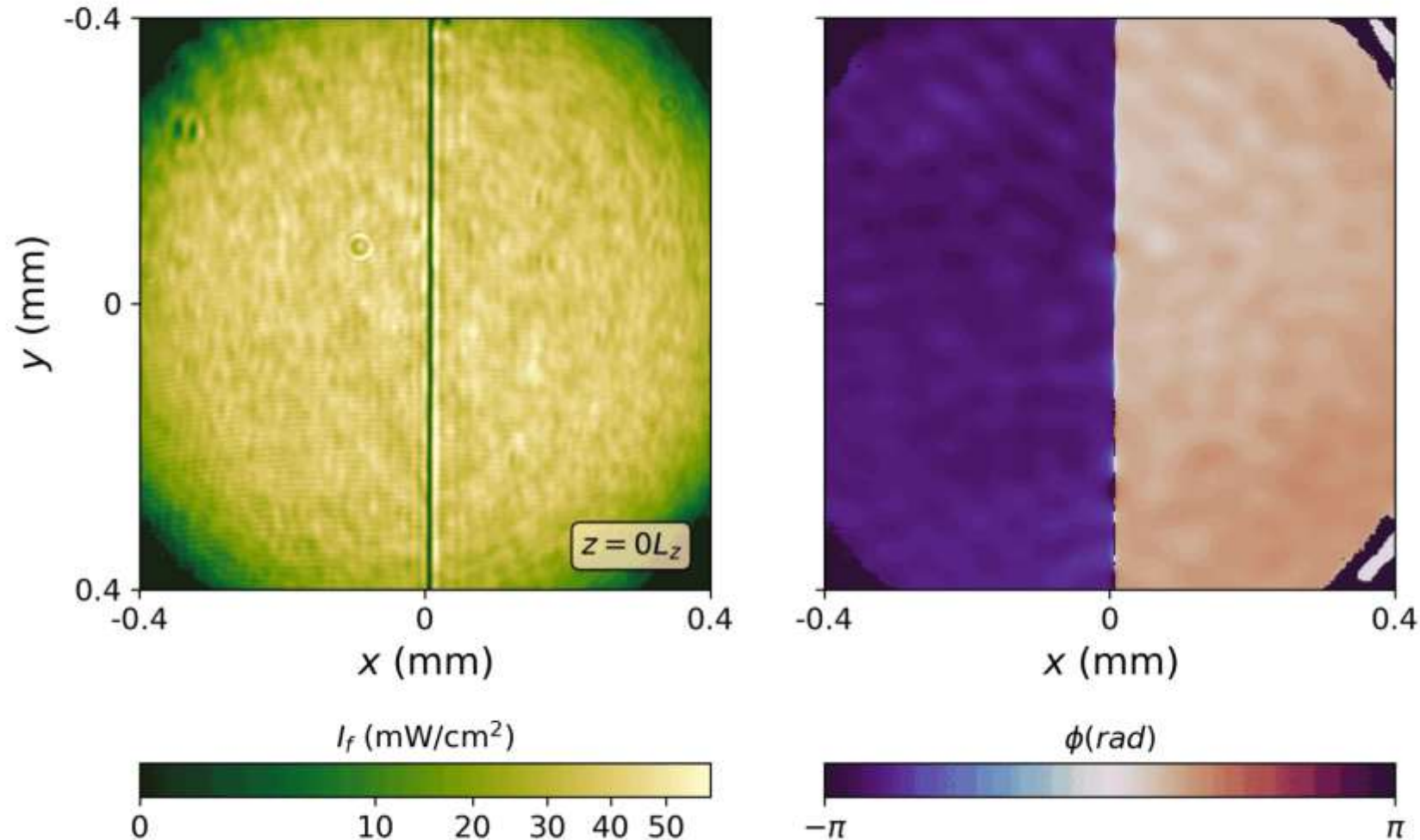
**Figure 21** – Feedback loop of the input state.



# Digital Feedback Loop in Fluids of Light

Experimental results: Dark Soliton decay and shock waves expansion

Feedback Loop for 5 passages!



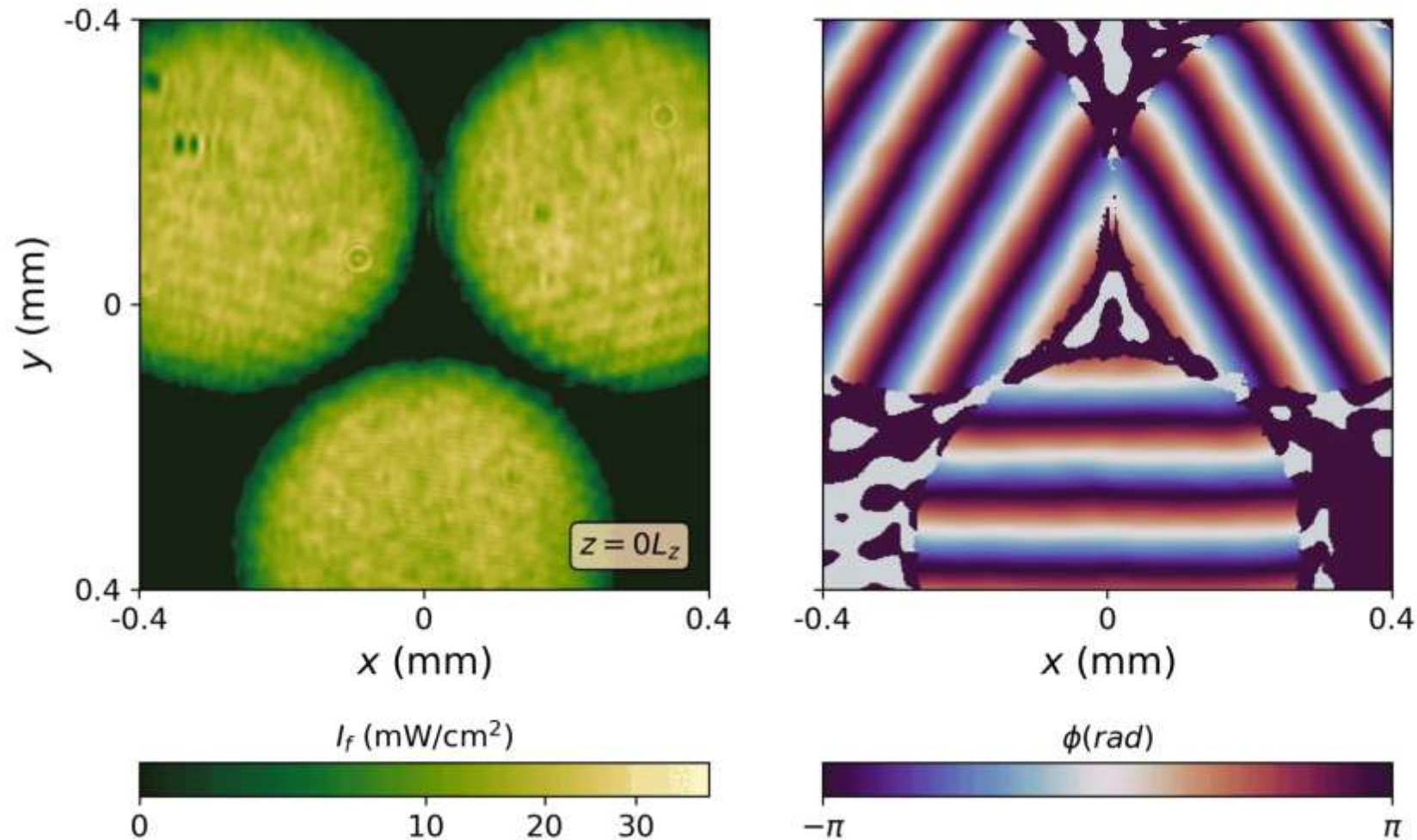
**Figure 22** – Dark soliton decay and snake instability.

Tiago D Ferreira et al. PhysRevLett.133.243802 (2024)

# Digital Feedback Loop in Fluids of Light

Experimental results: Collision between three flat-top states

Feedback Loop for 6 passages!



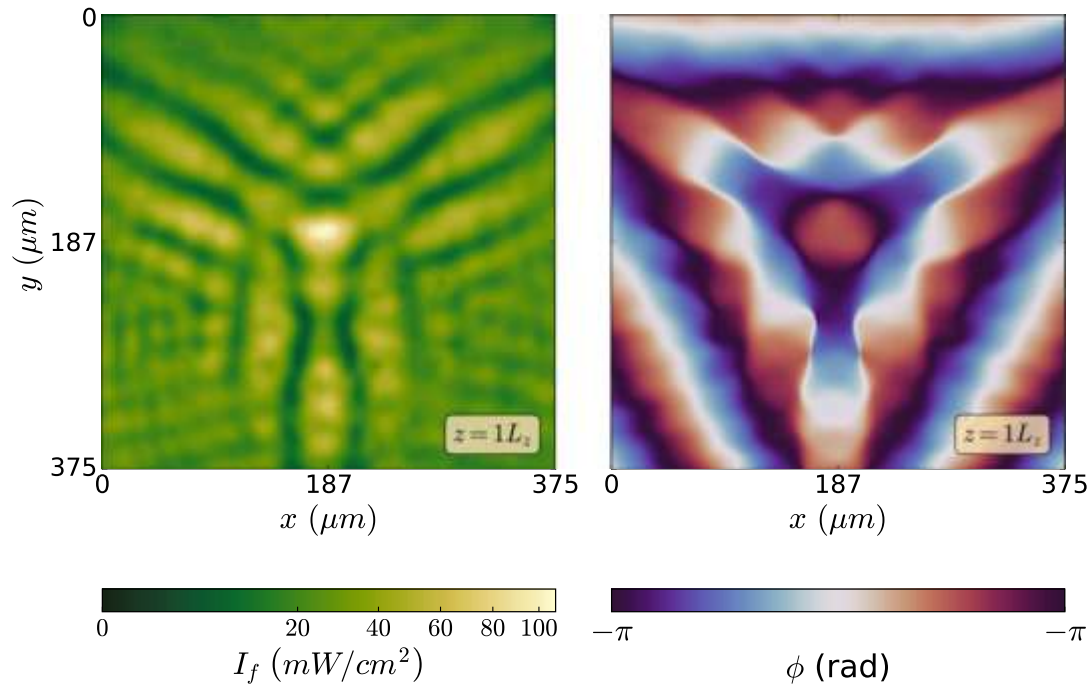
**Figure 23** – Collision dynamics between 3 flat top beams.

Tiago D Ferreira et al. PhysRevLett.133.243802 (2024)

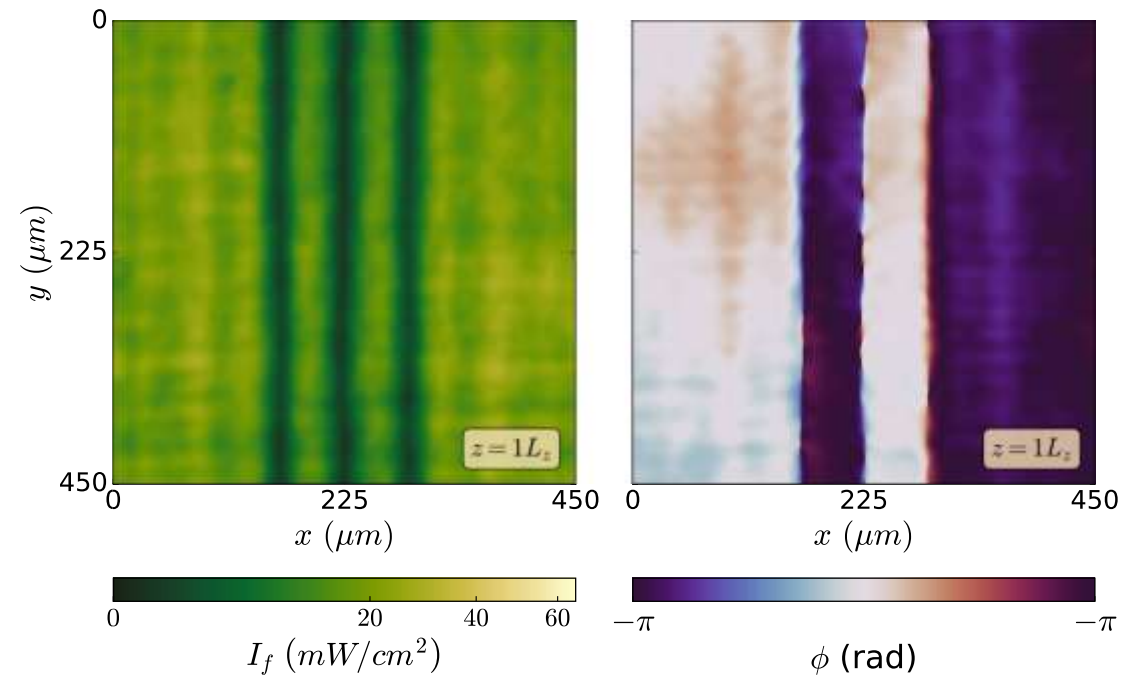
# Digital Feedback Loop in Fluids of Light

Experimental results: Other preliminary results

**Exploration** of more **complex** and **non-trivial** configurations.



**Figure 24** – Dynamics for the vortex formation and vortex collision with sound emission



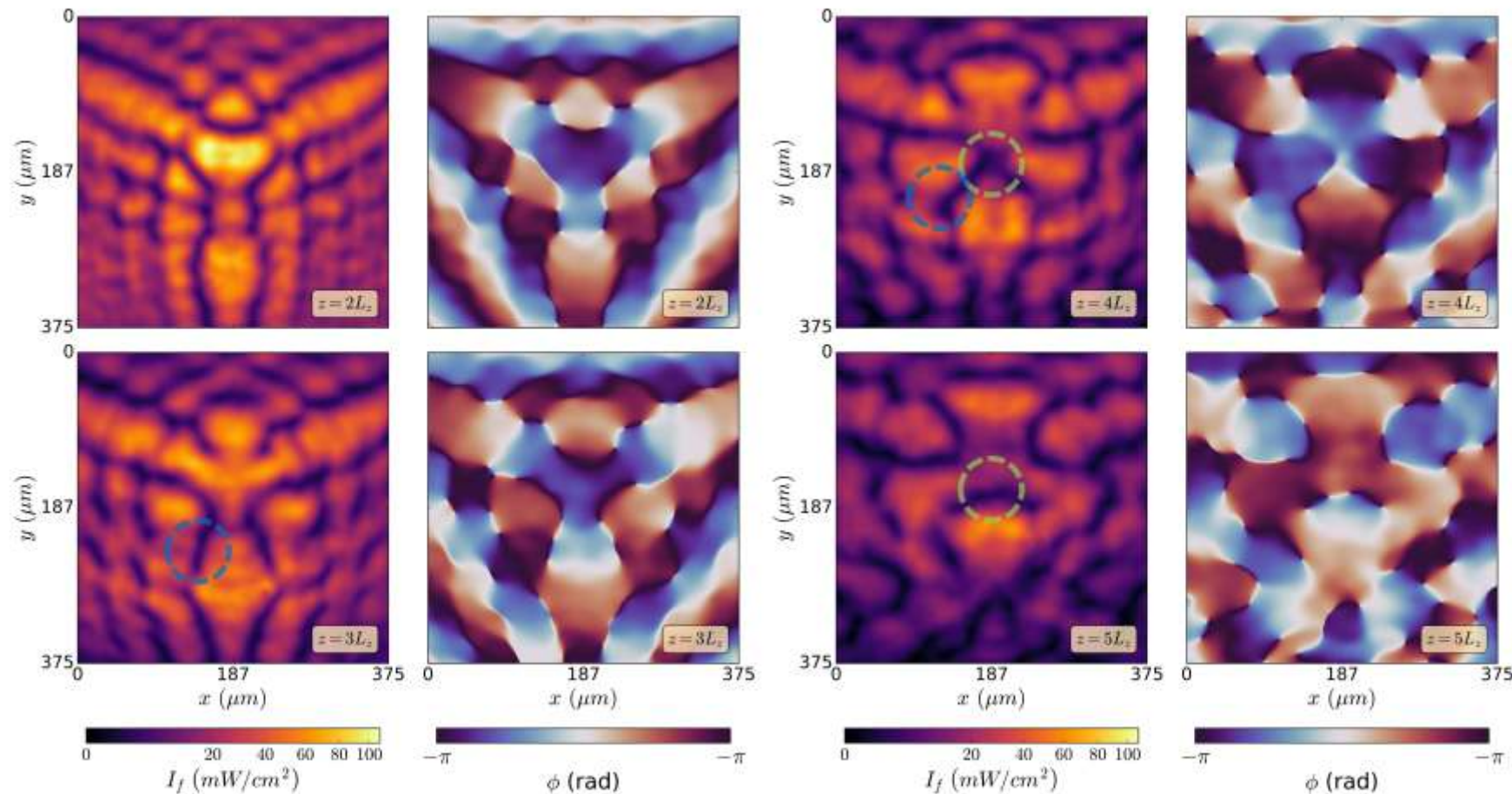
**Figure 25** – Dynamics for the decay of three dark solitons.



# Digital Feedback Loop in Fluids of Light

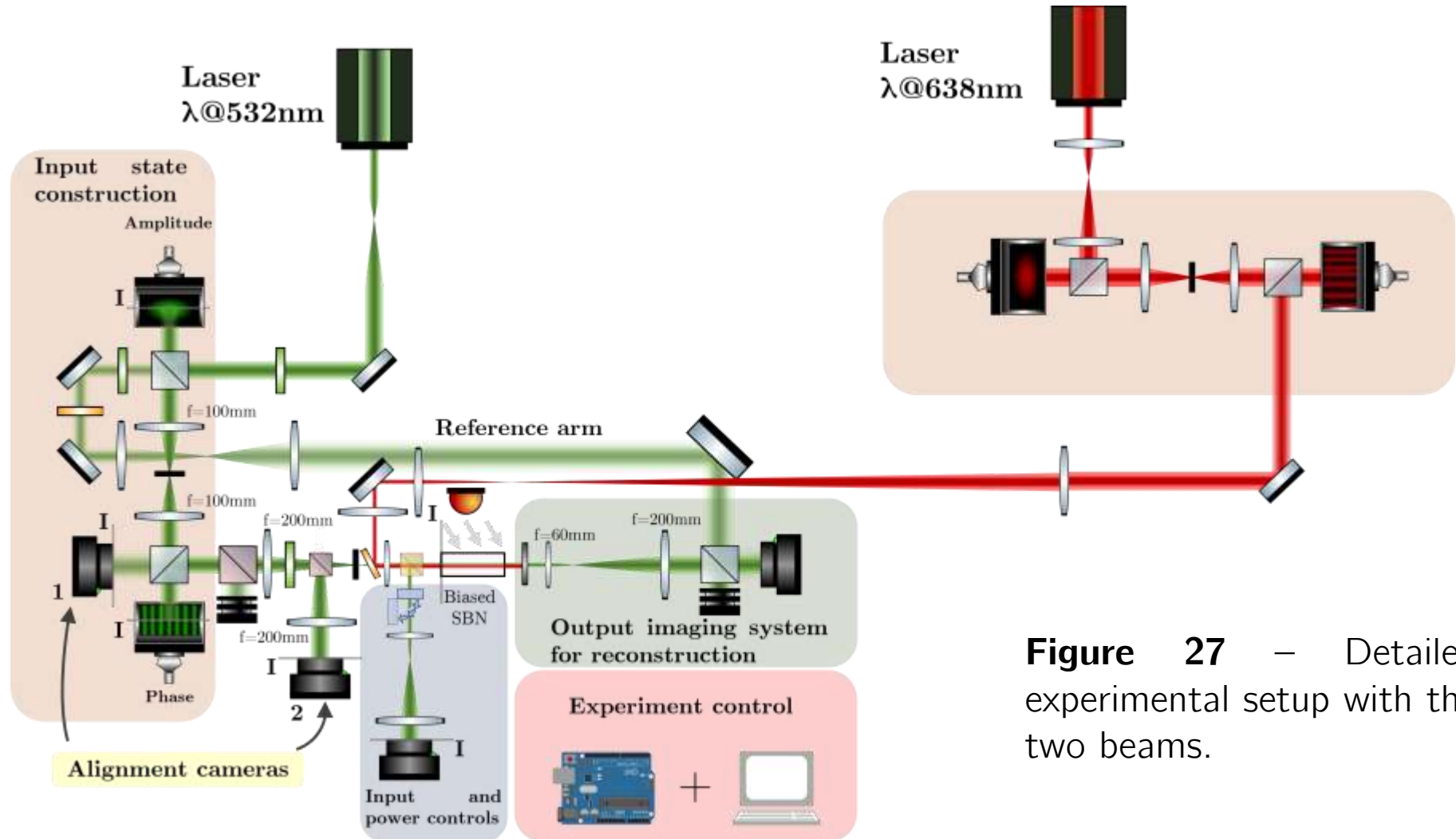
Experimental results: Other preliminary results

**Exploration** of more **complex** and **non-trivial** configurations.



**Figure 26** – Vortex collision and generation.

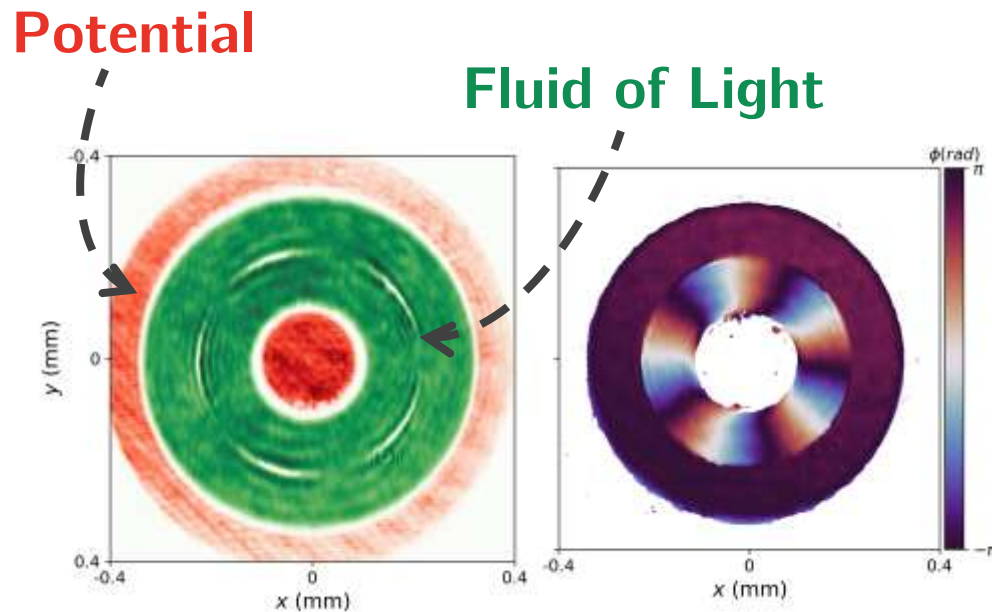
# Digital Feedback Loop in Fluids of Light with two beams



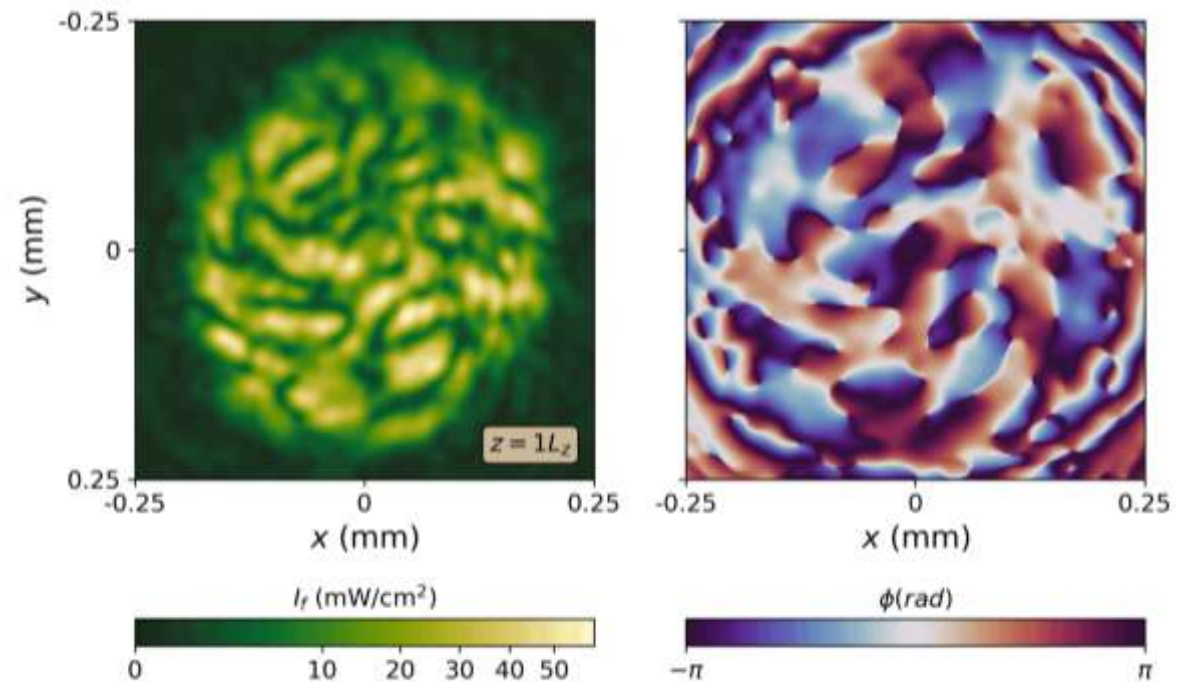
**Figure 27** – Detailed experimental setup with the two beams.

# Digital Feedback Loop in Fluids of Light with two beams

Preliminary results: Confinement potentials!



**Figure 28** – Intricate confinement potentials

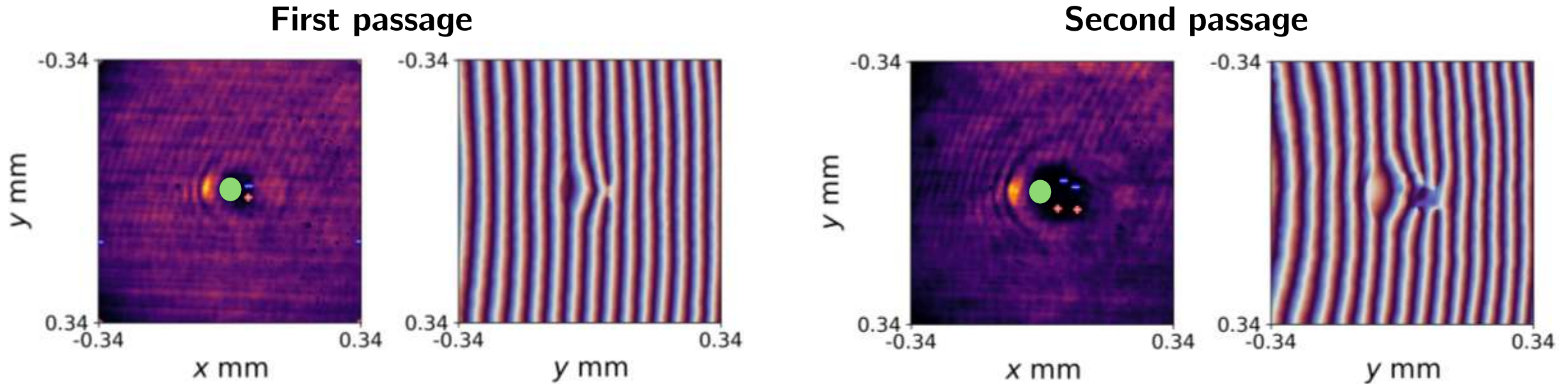


**Figure 29** – Turbulent dynamics in a confined configuration.

# Digital Feedback Loop in Fluids of Light with two beams

Preliminary results: **Fluid** with an **optical defect**

● → Optical defect

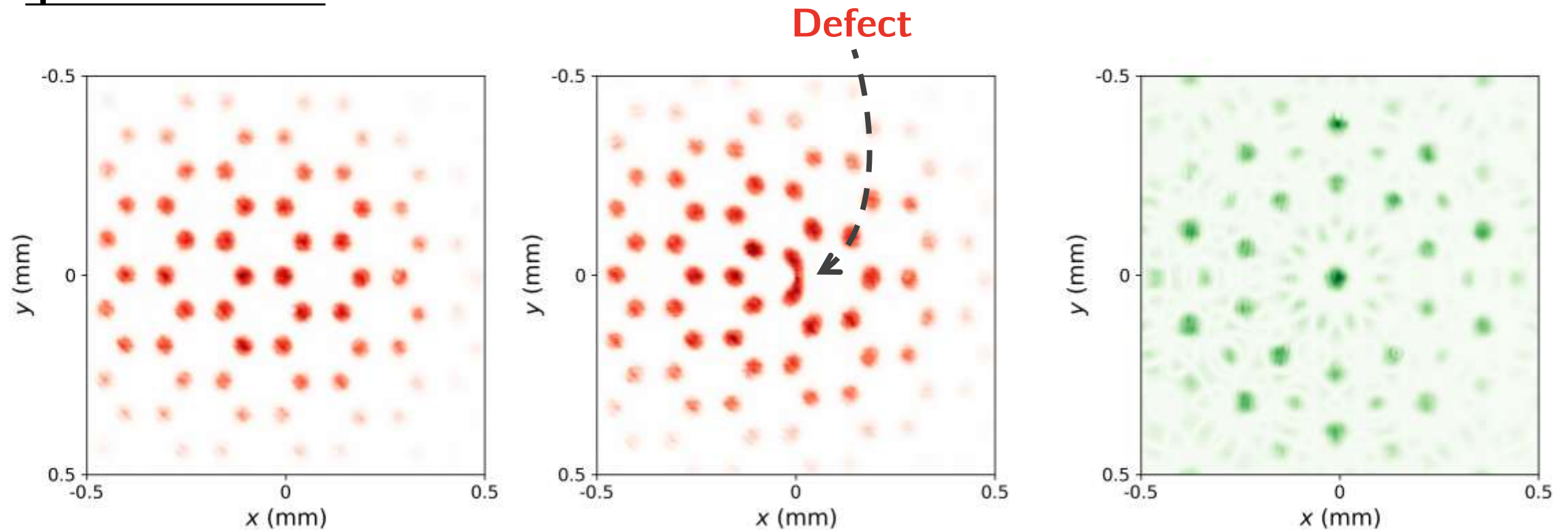


**Figure 30** – Feedback loop with two beams: vortex nucleation.



# Digital Feedback Loop in Fluids of Light with two beams

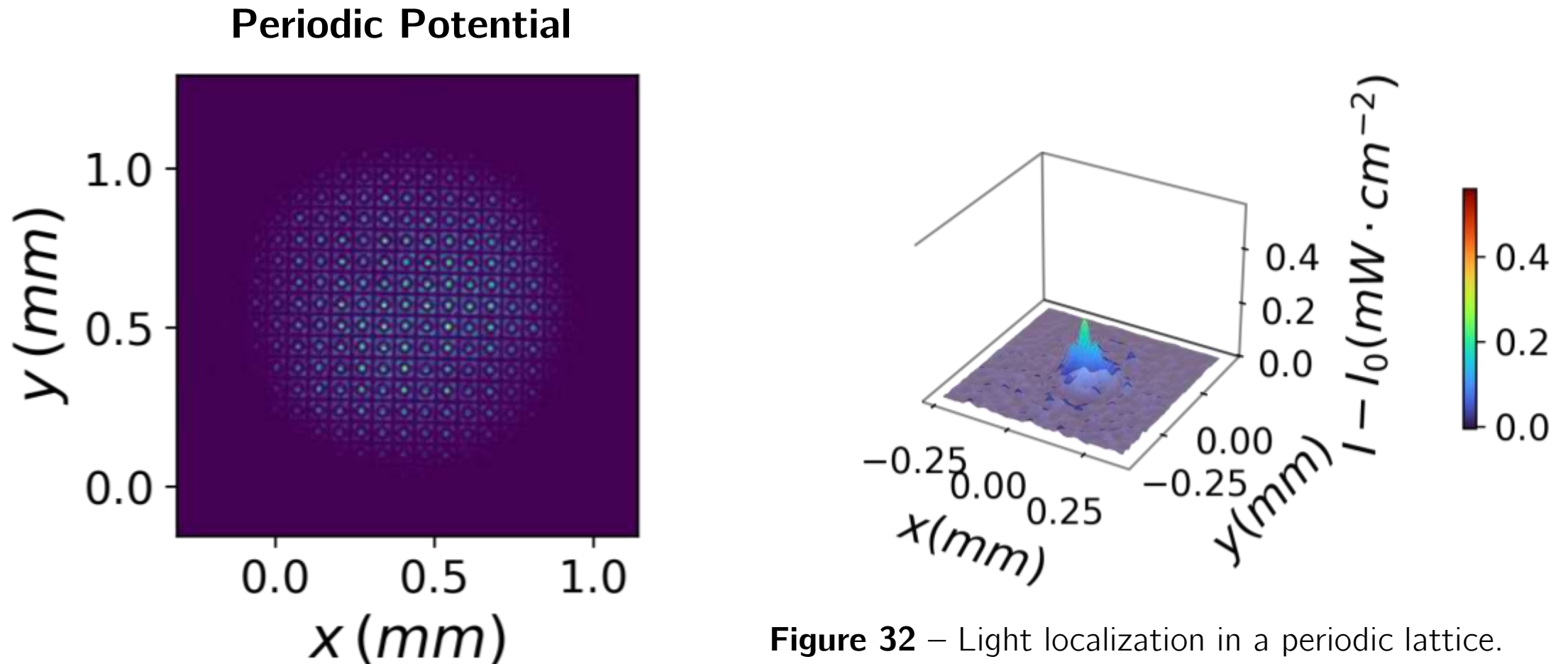
We also working towards topological quantum fluids and **analogue simulations of quantum matter**.



**Figure 31** – Topological potentials and quasi-crystal structures.

# Digital Feedback Loop in Fluids of Light with two beams

Preliminary results: **Light localization**: the **crystal** is in the **focusing regime**



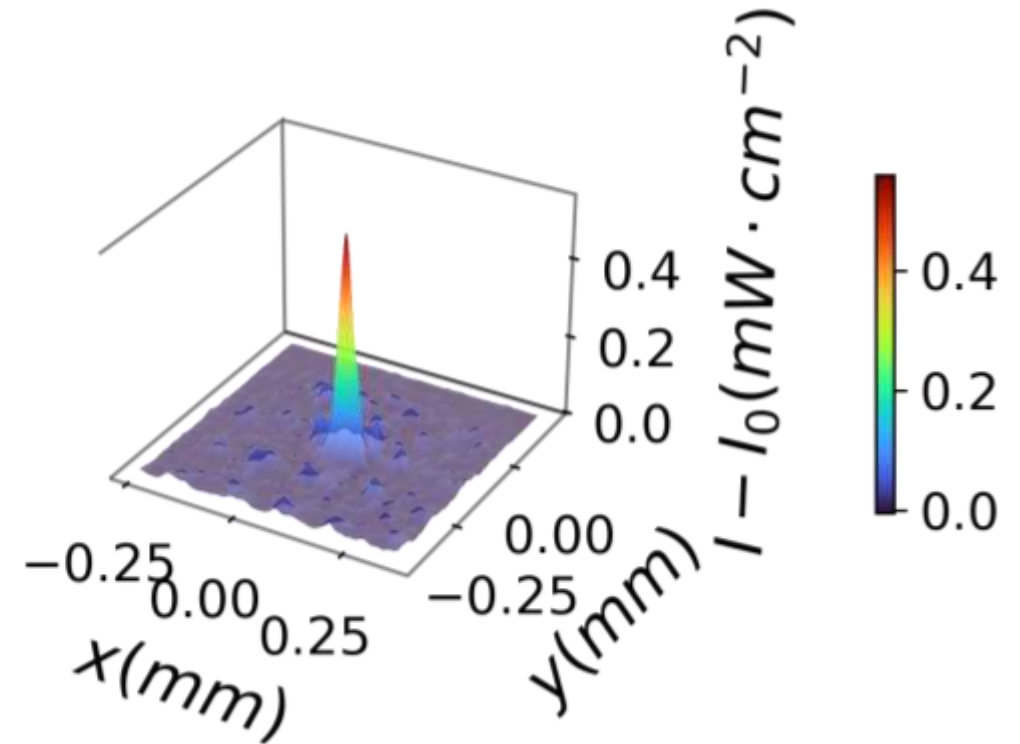
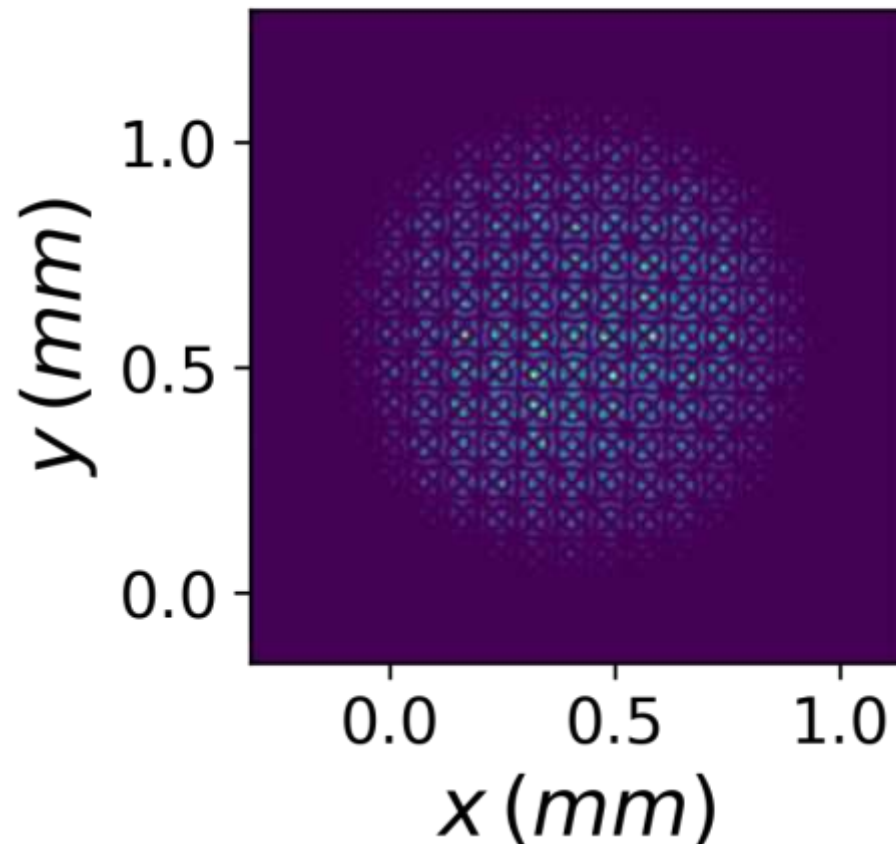
**Figure 32** – Light localization in a periodic lattice.



# Digital Feedback Loop in Fluids of Light with two beams

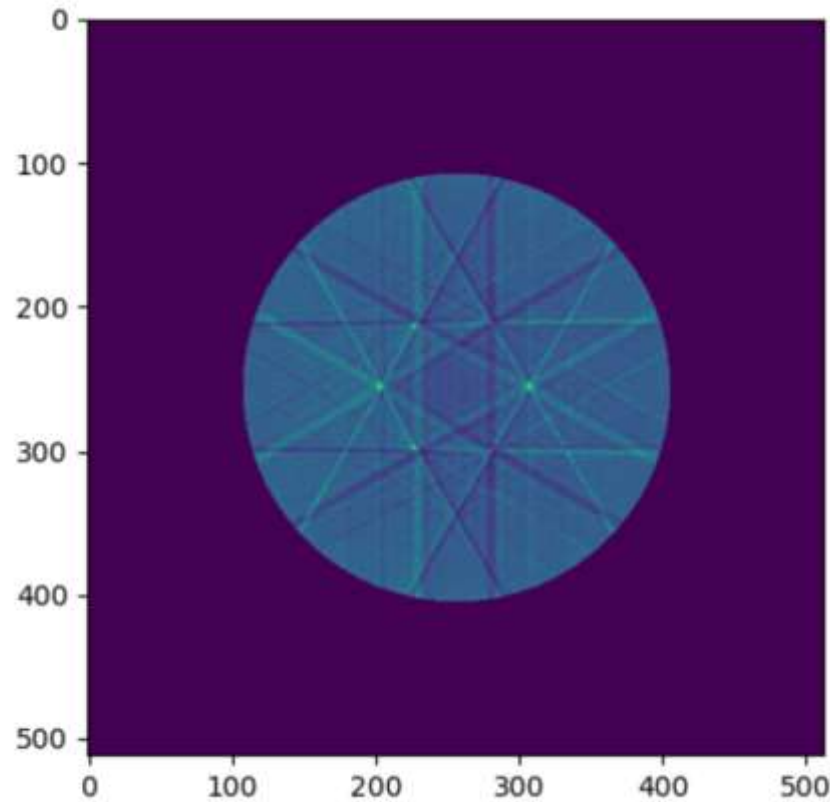
Preliminary results: **Light localization**: the **crystal** is in the **focusing regime**

Aperiodic Potential

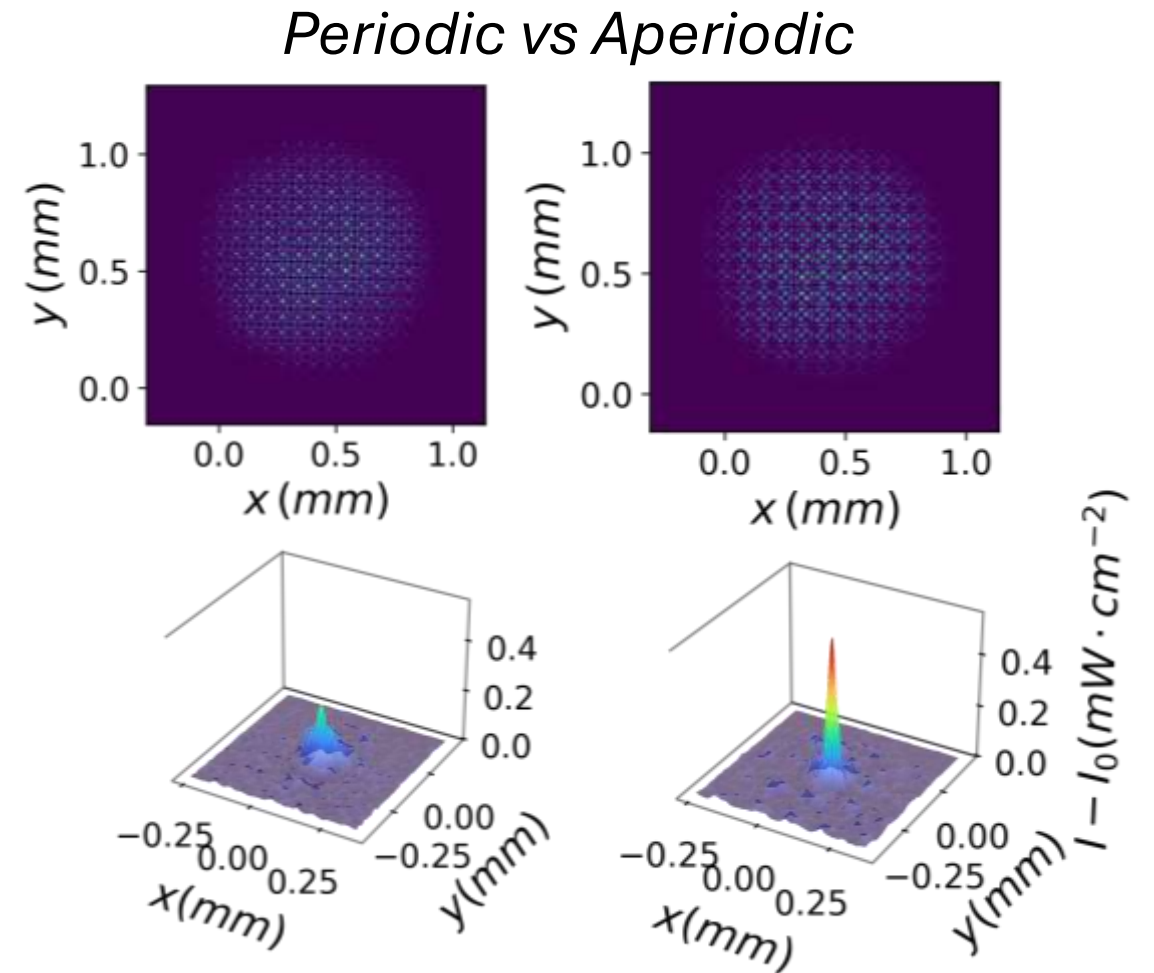


**Figure 33** – Light localization in an aperiodic lattice.

# Digital Feedback Loop in Fluids of Light with two beams



**Figure 34** – Numerical simulation of Brillouin zone measurement.

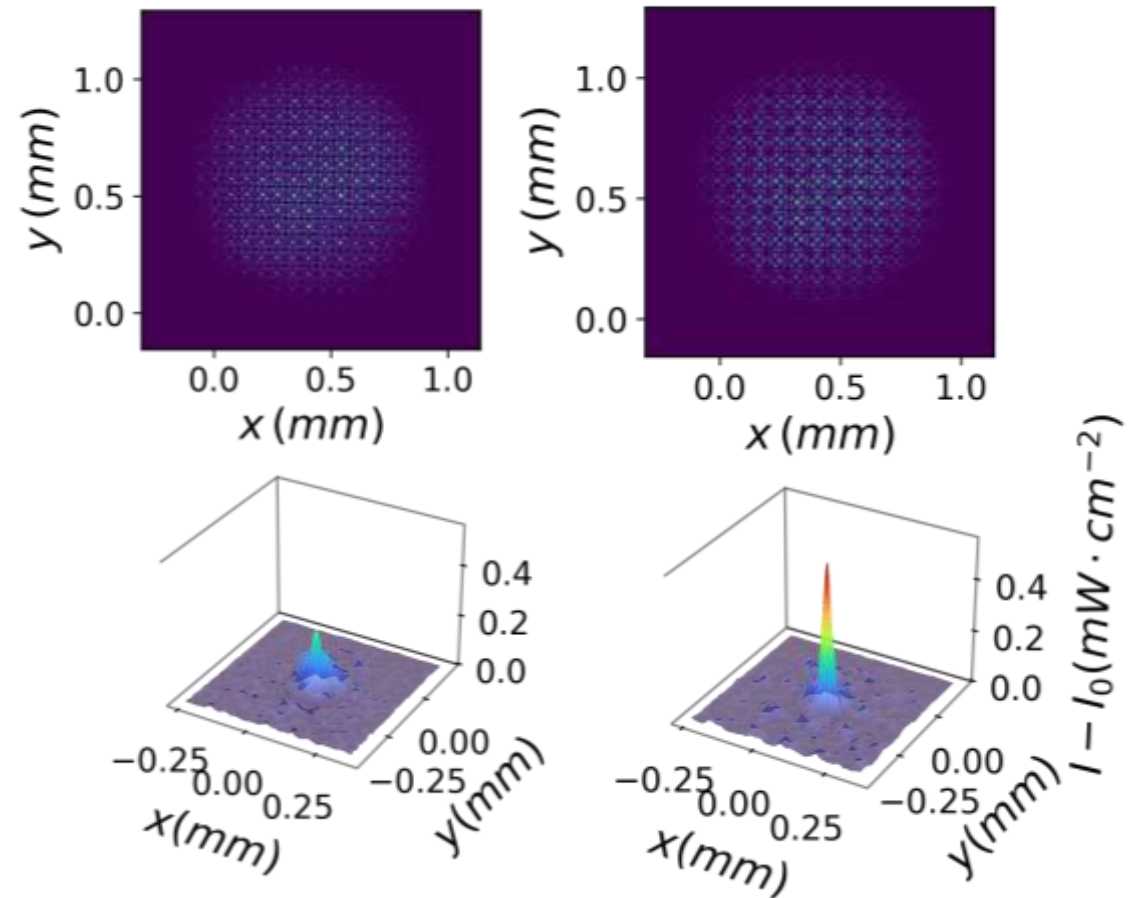


# Digital Feedback Loop in Fluids of Light with two beams

**Open question 1:** Can we use these analogue systems to speedup calculations of Brillouin-zones and then band structures?

**Open question 2:** Can we use these systems to support the engineering of new optical materials (e.g. photonic crystal fibers)?

*Periodic vs Aperiodic*



# Key Takeaways

- We have a **modular** and **versatile platform** for **paraxial fluids of light**.
- We are exploring **paraxial fluids of light** towards technological applications.
- The **Digital Feedback Loop** works; however, it is **necessary** to **map** its potential **applications** as well as its **limitations**.
- There is a lot of **room** for **improvements**, so we are **open** to **new ideas** and **collaborations**.

Our Website

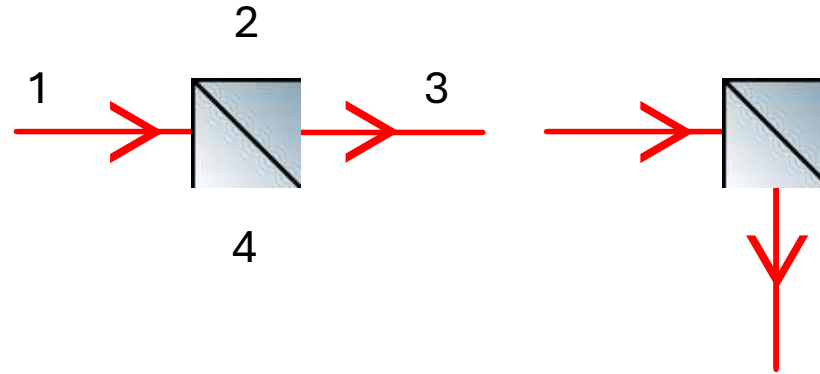


Quantum, Optics and Spectroscopy team,

Part of INESC TEC's Centre for Applied Photonics.

Light as a Versatile Platform for Analogue Physical  
Simulations and **Quantum Imaging Applications**

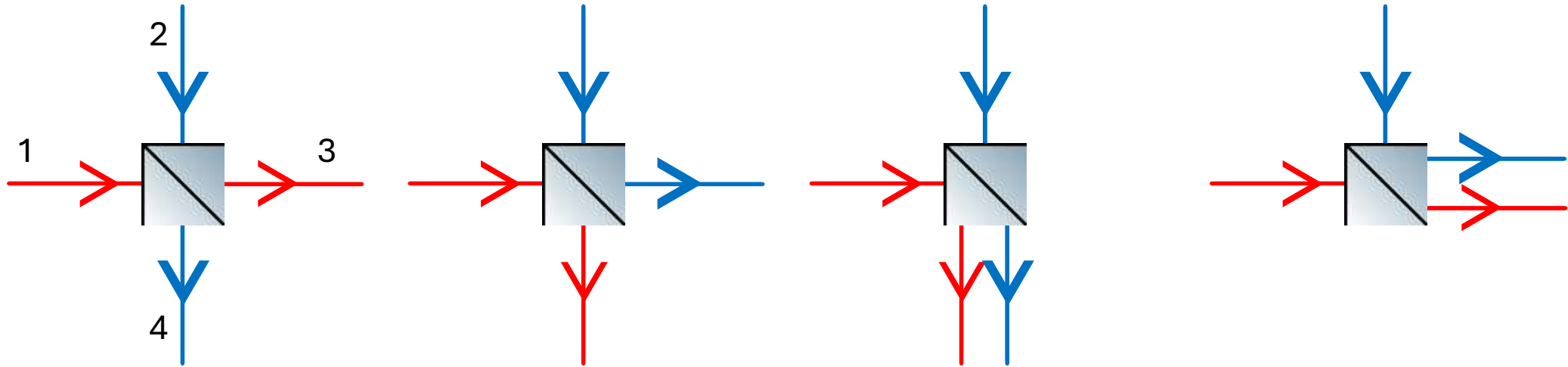
# Quantum Imaging: Probing the Fast-Axis Angle using Hong–Ou–Mandel Interference



$$|\psi\rangle = |1,0\rangle_{1,2} \Rightarrow |\psi\rangle = \frac{1}{\sqrt{2}} (|1,0\rangle_{3,4} - |0,1\rangle_{3,4})$$



# Quantum Imaging: Probing the Fast-Axis Angle using Hong–Ou–Mandel Interference



$$|\psi\rangle = |1,1^*\rangle_{1,2} \Rightarrow |\psi\rangle = \frac{1}{2} (|1,1^*\rangle_{3,4} - |1^*,1\rangle_{3,4} - |0,2\rangle_{3,4} + |2,0\rangle_{3,4})$$

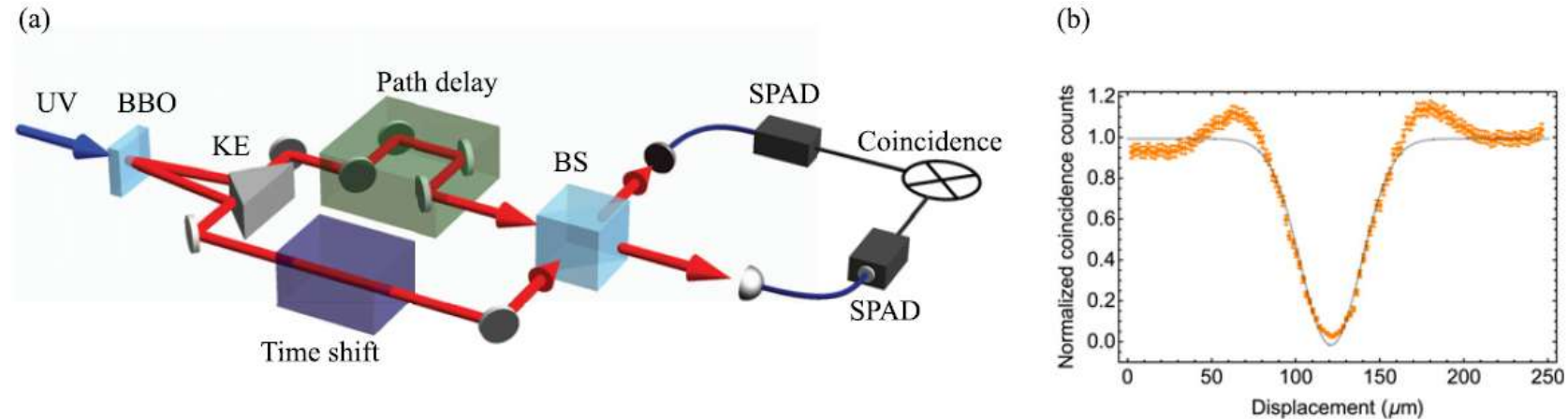
If the **photons** are **indistinguishable**, these terms **cancel out**

# Quantum Imaging: Probing the Fast-Axis Angle using Hong–Ou–Mandel Interference

**Fourth-order interference** via **photon bunching** at a 50:50 beam splitter.

No phase stability required; **enables low-noise imaging**.

**Dip depth, visibility, or position** reveal photon properties such as **delay** or **polarization alignment**.



**Figure 35** – Hong-Ou-Mandel Scheme.

Bouchard et al, DOI: [10.1088/1361-6633/abcd7a](https://doi.org/10.1088/1361-6633/abcd7a)

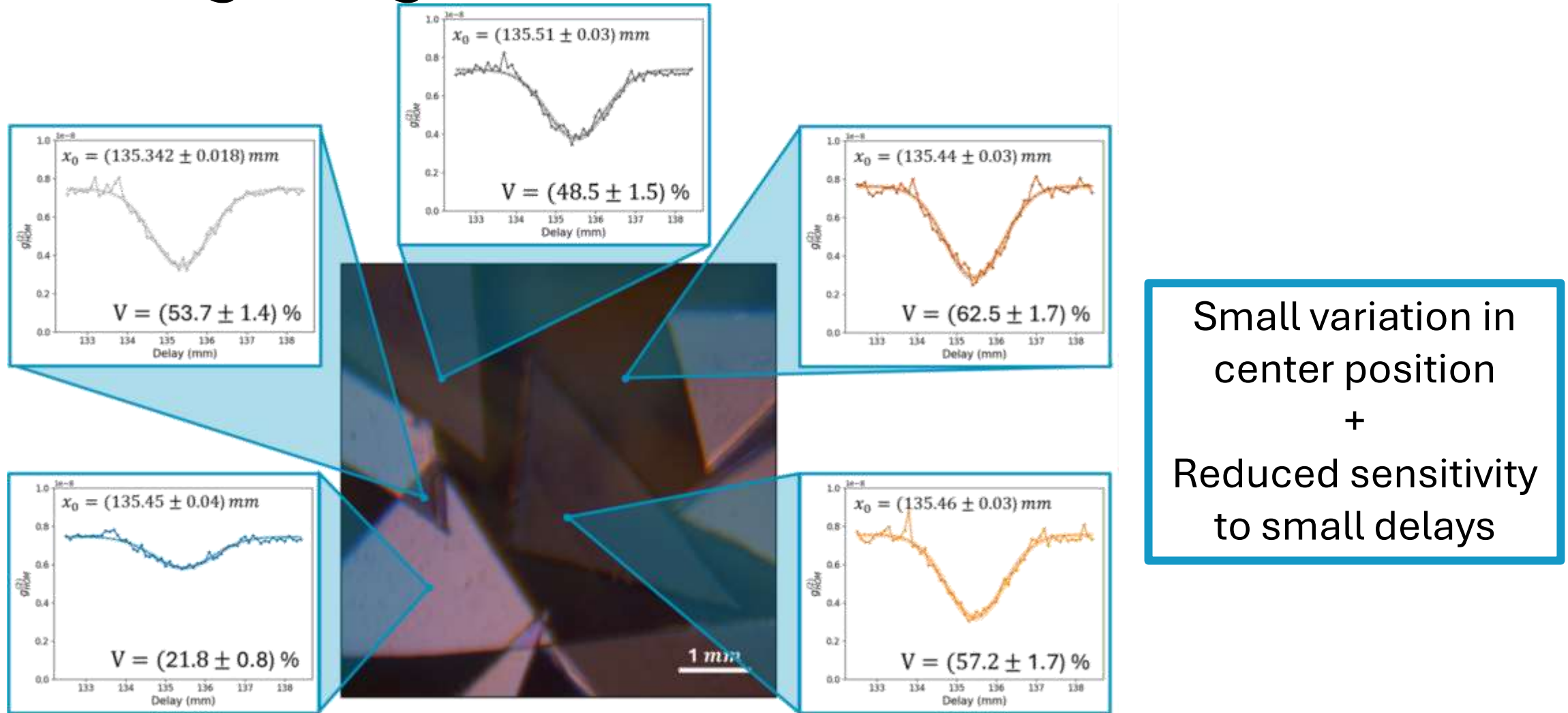
# Quantum Imaging: Probing the Fast-Axis Angle using Hong–Ou–Mandel Interference

**HOM dip** usually **depends** on **multiple factors**, making individual effects hard to isolate.

Example: **birefringent samples** can **change** both **photon delay** and **polarization alignment**.

We use a **narrowband single-photon source** with **mm-wide HOM dips**, making the setup **mainly sensitive to polarization alignment**.

# Quantum Imaging: Probing the Fast-Axis Angle using Hong–Ou–Mandel Interference



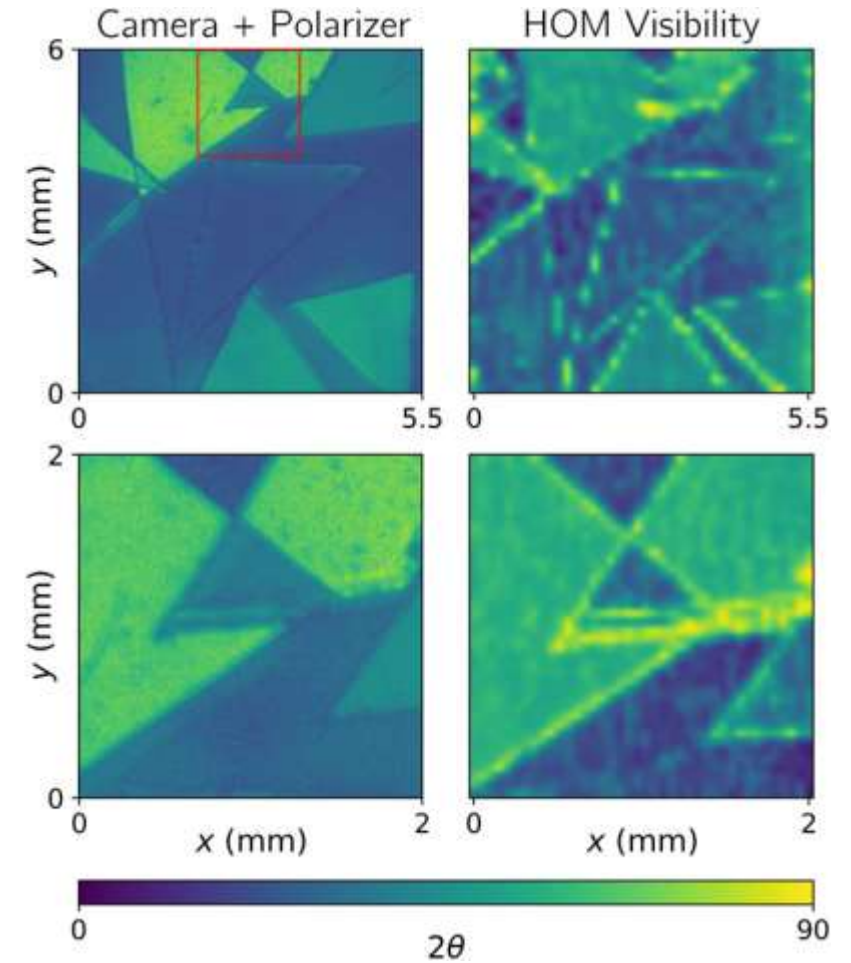
**Figure 36** – Hong-Ou-Mandel curves measured at different locations of a birefringent sample

# Quantum Imaging: Probing the Fast-Axis Angle using Hong–Ou–Mandel Interference

Used **Fisher information** to **estimate** the **fast-axis angle** in birefringent samples.

Found the **dip minimum** to be the **most informative operating point**.

Enables **polarization-sensitive imaging without absolute phase stability**, using **low-power, non-damaging light**.



**Figure 37** – Experimental results of the fast-axis angle obtained using classical and the HOM approach.

# Quantum Imaging: Imaging with Undetected Photons

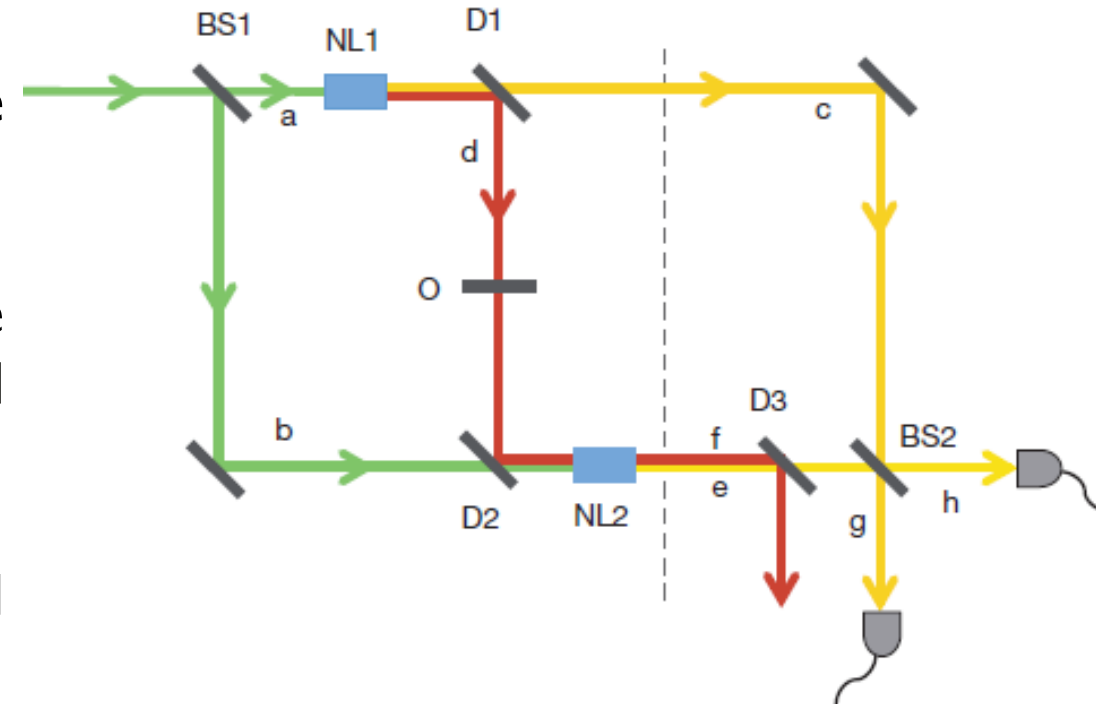
We are also exploring a **different approach** for **quantum imaging** that uses **undetected photons**

This technique allows to **probe** in the **IR regime** **without using IR sources and detectors.**

The sample is placed on the **idler arm**, and those **photons** are **never measured**. Only the **signal photons** are measured with a commercial CMOs.

This is possible due to **quantum induced coherence.**

Crystal **temperature** and **polling grating** allows to **select different wavelengths** - **Hyperspectral**



**Figure 38** – Experimental scheme for imaging with undetected photons.

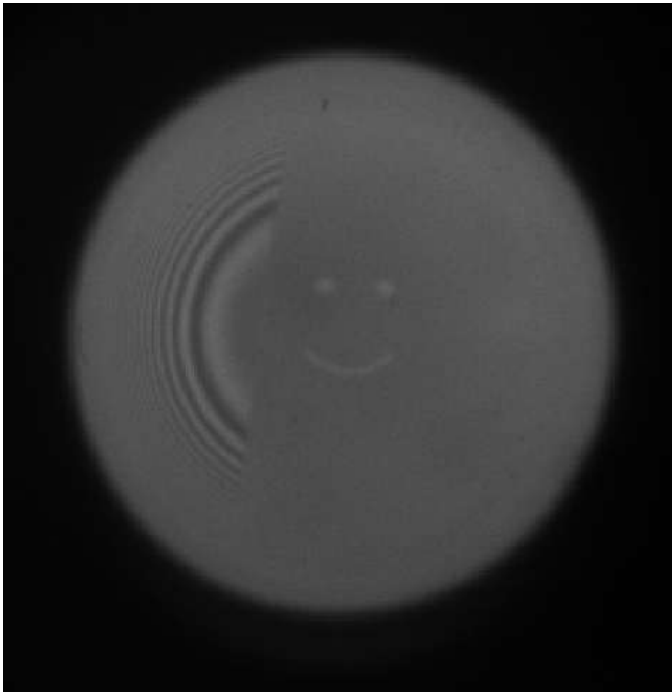
Lemos, G. et al. Nature 512, 409–412 (2014).



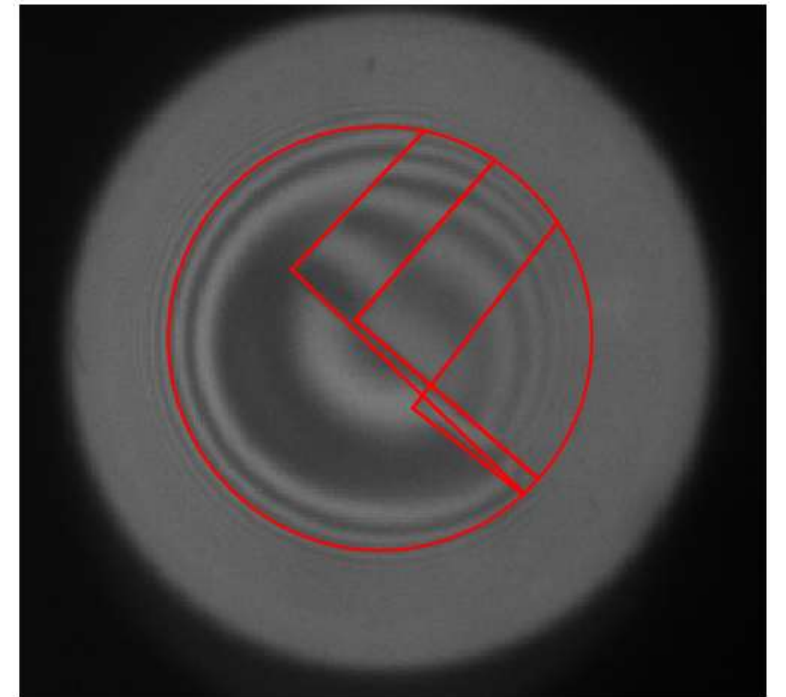
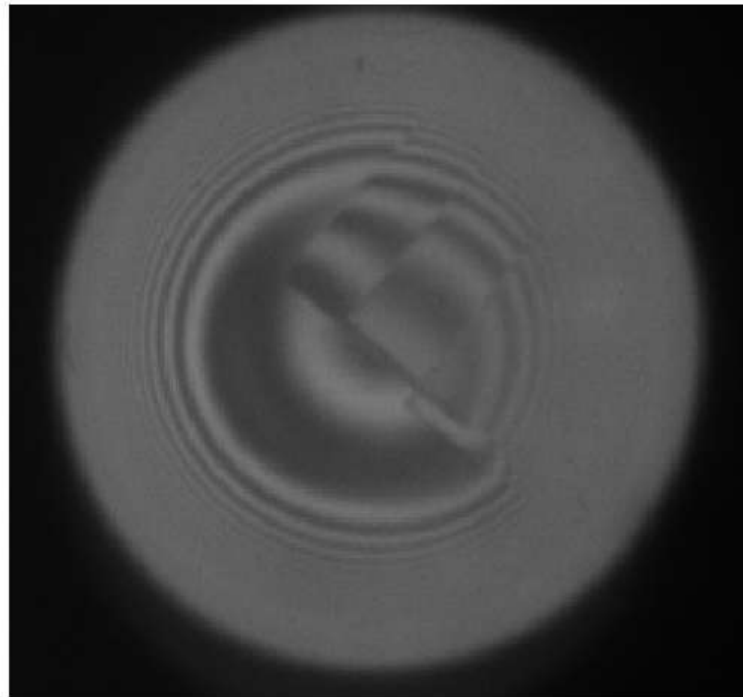
# Quantum Imaging: Imaging with Undetected Photons

Preliminary results: We are **probing  $\lambda \sim 1550 \text{ nm}$**  but the **measuring photons are  $\lambda \sim 810 \text{ nm}$**

T=1 vs T=0



Path Delay changes



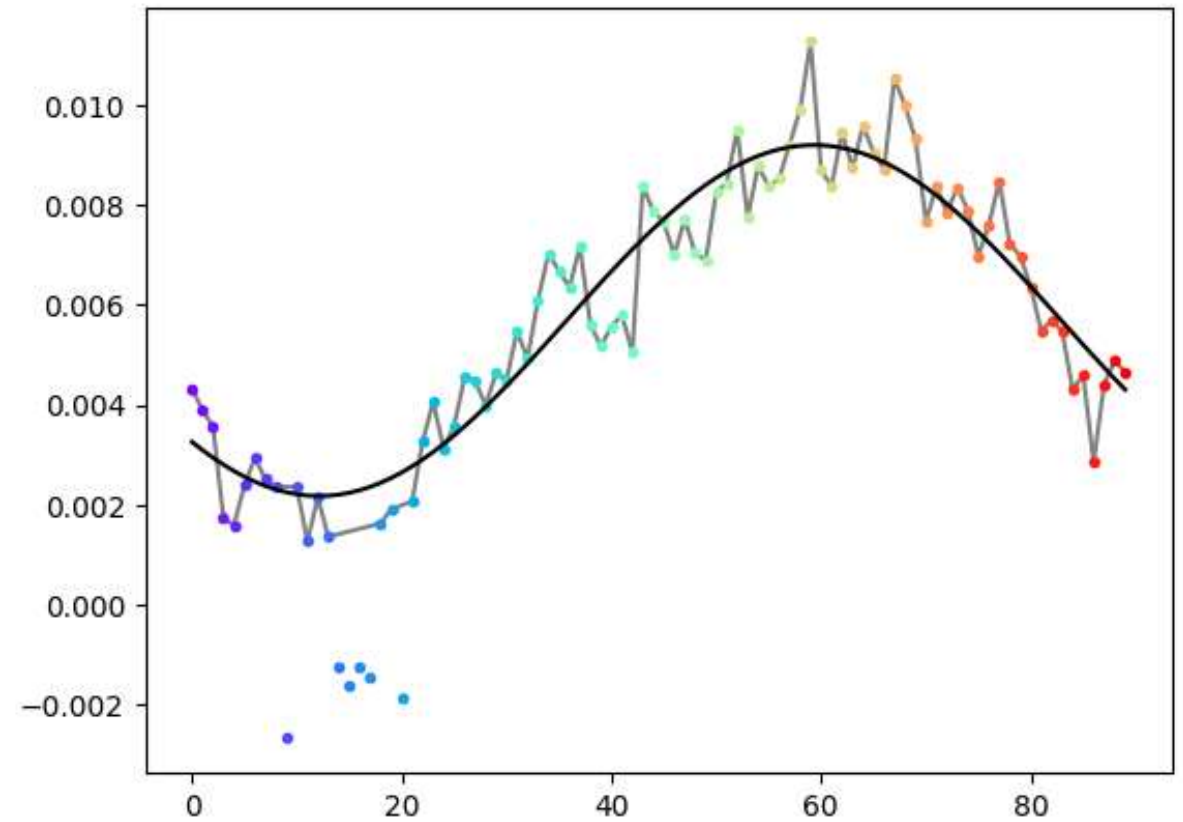
**Figure 39** – Amplitude and phase imaging, with the idler around the 1550nm.

# Quantum Imaging: Imaging with Undetected Photons

Preliminary results:

We are also exploring the possibility of using **imaging with undetected photons** for **birefringence imaging**.

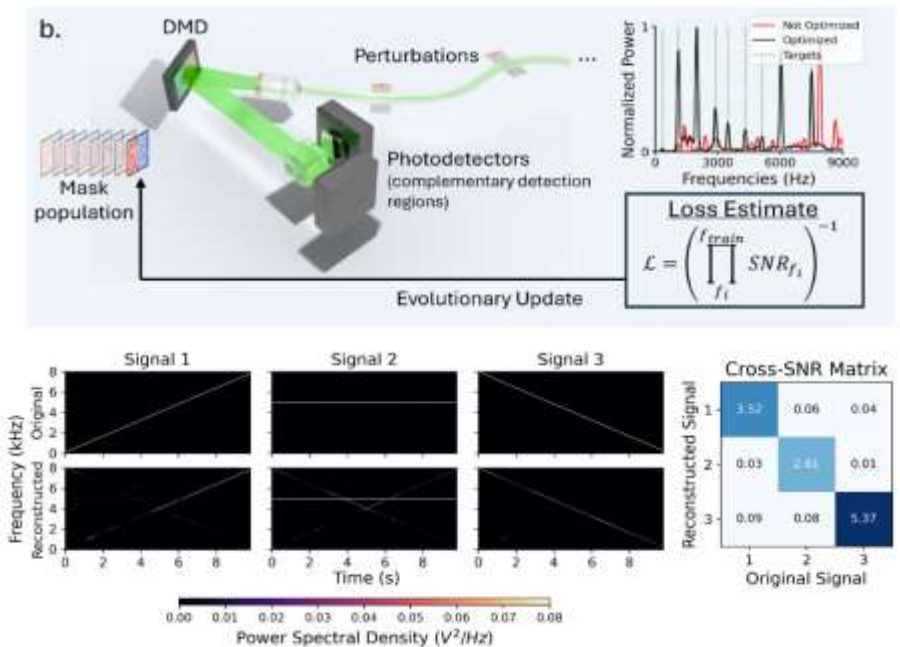
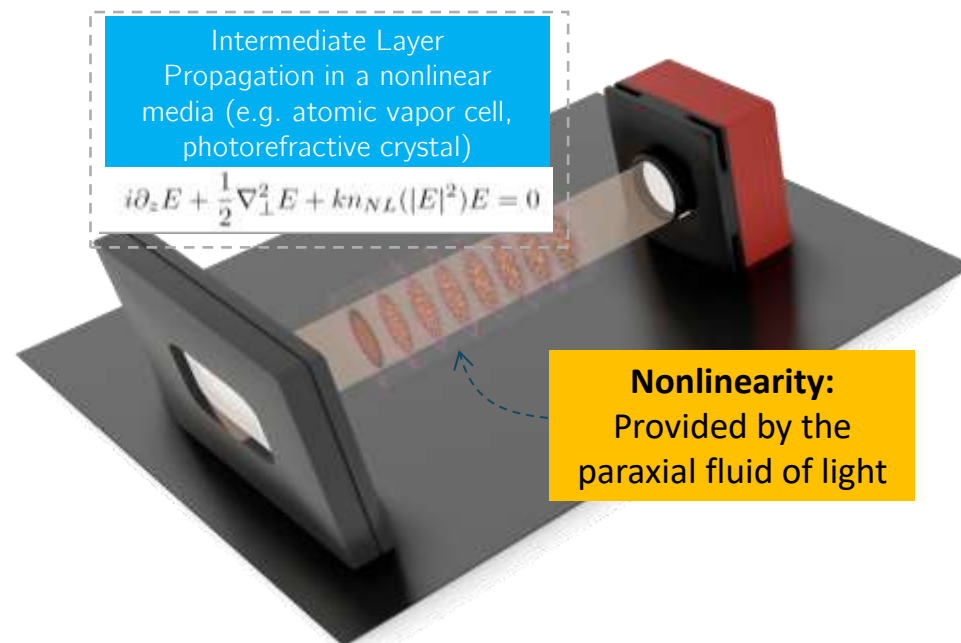
**Towards Hyperspectral Birefringence Imaging.**



**Figure 40** – Amplitude of the cosine fit of the interference bands vs angle of the quarter wave plate

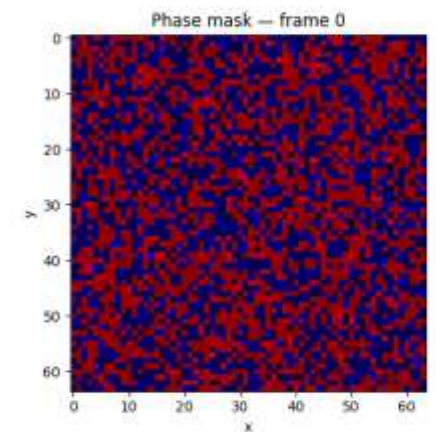
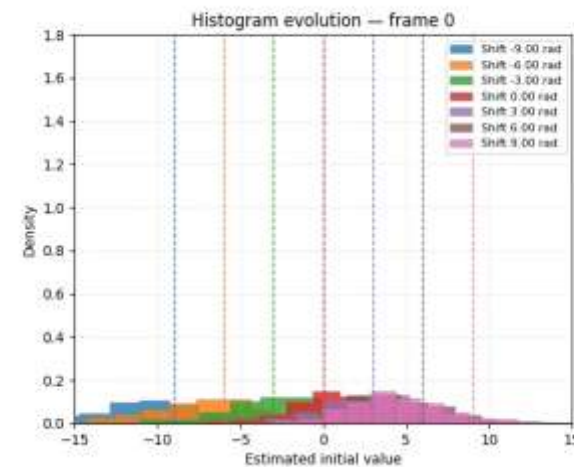
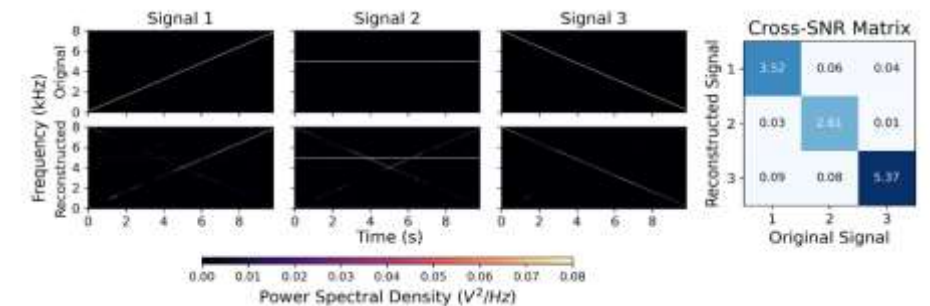
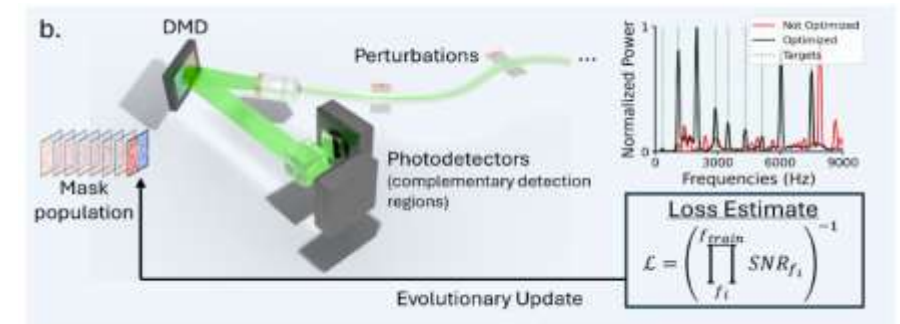
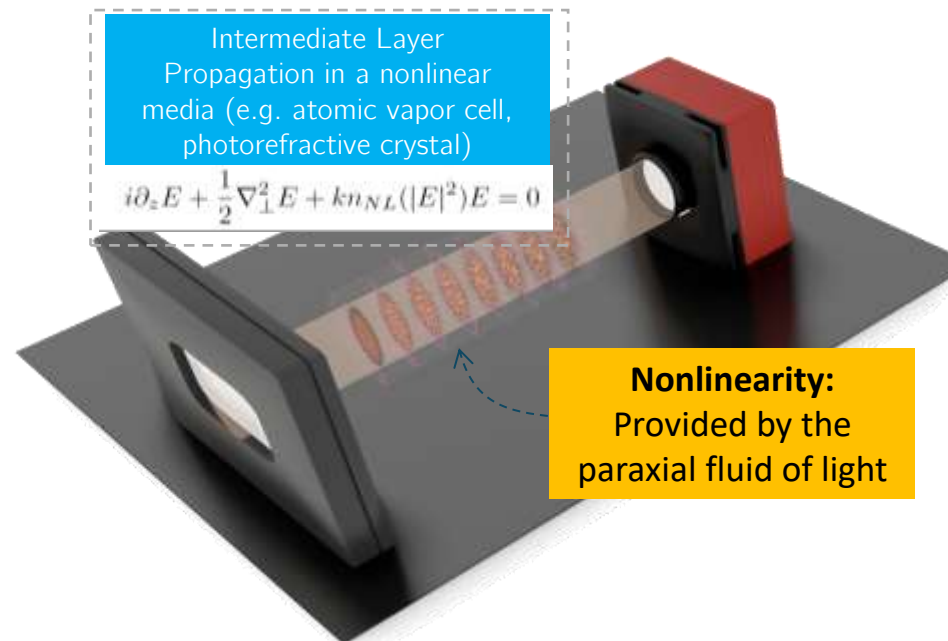
# Optical Processing and Computing

- **Extreme Learning Machines, Reservoir Computing** (with paraxial fluids of light).
- **Optical Processing.**
- **In Situ Learning**



# Optical Processing and Computing

- **Extreme Learning Machines, Reservoir Computing** (with paraxial fluids of light).
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## Our Website



## Acknowledgments

This work is co-financed by Component 5 - Capitalization and Business Innovation, integrated in the Resilience Dimension of the Recovery and Resilience Plan within the scope of the Recovery and Resilience Mechanism (MRR) of the European Union (EU), framed in the Next Generation EU, for the period 2021 - 2026, within project HfPT, with reference 41. This work was financed by the National Funds through the FCT—Fundação para a Ciência e a Tecnologia, I.P. (Portuguese Foundation for Science and Technology) within the project HyperQulm, with reference 2024.17596.PEX (<https://doi.org/10.54499/2024.17596.PEX>)



**Thank you for your attention**





# Light as a Versatile Platform for Analogue Physical Simulations and Quantum Imaging Applications

Armchair: Tiago D. Ferreira

Workshop: Quantum-SOPHA and Armchairs, Peyresq, 2026



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Quantum, Optics and Spectroscopy team

